Book of abstracts

Biannual DASTS conference

DASTS24: Elegies of waste, surplus, and excess

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Danish Association for Science and Technology Studies

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Open panel A

Data issues and methods
Chair: Brit Ross Winthereik

This panel invites presentations by researchers interested in discussing effects of the figure of 'the digital citizen' as well as how a reorientation of STS might help nuance it as a policy concept.

Our point of departure is that 'the digital citizen' as a policy concept causes many problems that needs fixing to avoid further exclusion and breaches of fundamental human rights in digitalized states. We believe STS can help by describing under what circumstances infrastructures meant to provide public services become frictional.

We are especially interested in the elaborating the concept of human experience as it has been used in STS to also encompass digital infrastructures as environment for diverse livelihoods.

We relate to the conference theme especially through the notion of excess, which we consider all the aspect of citizens that cannot be contained by the digital technologies in use for accessing public services. Thus, by problematizing a taken-for-granted concept in the digital state and by reports from attempts to navigate the complex passage between state authorities and citizens (or non-citizens), we hope to receive abstracts that can help us understand:

- The experiences of citizens and non-citizens of digitalization through empirical examples
- How such empirical knowledge about citizen and non-citizen experiences of digitalization may lead to policy contributions/improvement of current policy
- Conceptual innovation around (non)citizen experiences of digital infrastructures in the digital state
- Methodological innovation within digital public services to address the current lack of representation (co-creation, artistic methods, co-laboring etc.)

Chiara Carboni and Nanna Bonde Thylstrup ***Clinical data residues: On data quality and its implications for care provision***

Last year, a leaked Google memo disclosed the company's concerns around Open-Source large language models performing better than proprietary ones. Google computer scientists identify the degree of dataset curation, rather than the size of the dataset, as a prime factor that enables generating well-performing models in reasonably little time. On a similar line, but in a very different domain, data(set) quality is also emerging as a matter of concern for machine learning in healthcare. Earlier discourse on clinical artificial intelligence identified the healthcare sector as an ideal domain for development and application for machine learning models by virtue of the

huge amount of data ‘routinely produced’ in clinical practice. Now, increasingly, developers and practitioners alike single out the poor quality of this routinely-produced data as an issue weighing down on current experiments in machine learning. Mirroring emergent interests in paradata, growing emphasis is being placed, for instance, on opening up the conditions of data generation through labelling. In this paper, we aim to speak to issues of data quality and their implications for care by thinking through clinical data as residues. Drawing from our previous ethnographies of datafied care practices, we conceptualise clinical data as the residues of irretrievable, situated encounters amongst patients, practitioners, and a host of technological materialities and infrastructures. Residues, in their slipperiness and defiance of control, complicate claims around improving data quality through labelling. Moreover, residues travel and are difficult to contain, and increasingly pervade patients’ and professionals’ sensemaking, as well as machine learning.

Anders Koed Madsen and Laura Kocksch ***What should a digital methods experiment do? Three proposals***

Digital methods have become mundane and comfortable companions in various parts of STS research, collaboration and teaching. Simultaneously, we observe a growth of experimental formats and engagements ranging from “Making & Doing” sessions to publications such as “Experimental Ethnography” or “Experimental Collaboration”. We propose it is worth exploring the connection between these two trends further by systematically asking what makes the use of digital methods experimental, or, put differently, what ought a digital methods experiment do?
One proposal is to judge digital methods by what kind of analytic purview they enable. In this talk, we analyse analytic strategies that are known as “experimental” in three traditions: Ethnomethodology, American Pragmatism, and Anthropology. Each of these traditions uses experiments/experimental practices to craft particular analytic situations that enable epistemic inquiries albeit with very different goals and tactics. We propose that digital methods can take inspiration from each when crafting data sprints and other collaborative formats.

When working with digital methods at TANTlab, we aim for a diverse set of analytic situations. No digital methods experiment follows ever one tradition or another (neither are the traditions independent or symmetric). But clarifying what we mean when we say we do digital methods experimentally will advance our vocabulary to describe digital methods practices in contemporary STS. A series of questions emerge, such as what is it that data and visualizations do in one type of experiment or another?

Who experiments in one tradition or another? How is an experiment materially equipped in one tradition or another?

Christopher Gad and Katinka Schyberg *Digital remediations of community?*

As an organization, The Evangelic-Lutheran Church in Denmark (ELCD) is a hybrid form. The church and its employees operate at the intersection between ’administration’ and ’faith’ (Aggerbeck et.al 2023). The ELCD, of which 75% of the Danish population are members, is commonly known as the ‘folk church’, obviously serving religious purposes, but it is also a secular state bureaucracy, managing the Danish name registry. The latest Danish government foundation document expresses the hope and vision that the church will play a central role in saving Denmark from a perceived crisis of community and ongoing erosion of trust in public institutions. Digitalization of the public sector in Denmark has been profound. This is arguably connected to this crisisis. And while experiments occurred during Corona, with e.g. online communion, the church has also been more resistant to digitalization than the rest of the public sector. Lively debates are ongoing about whether and how specific digital technologies afford and affect religious as well as secular community building. And also, whether the church’s secular purposes in wider society are appropriate in governing the church and delineating its purpose or not. In this presentation we will present some initial observations from a collective research project offering examples of how digital tools challenge and provoke discussions of existing practices and ‘secular’ socio-material basis for religious conduct. Eg. Discussion of what a digital organ or a digital book of hymns afford - versus their traditional counterparts.

Katja Balslev *Unraveling Complexities: Understanding Shame, Bullying, and Betrayed School Cultures through Actor-Network Theory*

The phenomena of shame, bullying, and betrayed school cultures present complex challenges, termed "wicked problems," requiring nuanced solutions. Actor-Network Theory (ANT) serves as an analytical tool to comprehend the complexities within these contexts. ANT views bullying, distress, and shame patterns as outcomes of elaborate networks involving both human and non-human actors like social media and digital technologies. Particularly, digital bullying poses unique challenges due to anonymity and internet exposure, worsening its harmful effects. By employing ANT, this research explores interconnectedness and movements within networks, examining factors such as physical school environments, cultural dynamics, and non-human elements. It challenges conventional

dichotomies, emphasizing the interaction between different actors to understand emotional dynamics and their influence on bullying. Through ANT, this study aims to unravel the complex connections fostering shame and bullying, advocating for environments promoting respect, tolerance, and empathy to address these issues.

Panel 3

Taking stock: Design and STS sensibilities
Chair: Tau Lenskjold

For more than a decade, diverse areas of design research have been taking a concerted interest in STS as a source of theoretical and methodological inspiration. STS, it can be argued, has always been concerned with design practices and processes, through which the social, political and technological are imagined, constructed and realised (Danholt & Halse 2016). This has led to investigations into specific design practices and domains, such as the design studio (Farias & Wilkie 2018) and user representations (Hyysalo, Jensen & Oudshoorn 2016), and to appropriation of design-based methods, such as the design prototype (Jiménez 2014). STS-based theoretical frameworks have in turn been put into use as analytical strategies across design research, from design history (Fallan 2010) and design culture (Kimbell 2012) to participatory design (Björgvinsson, Ehn & Hillgren 2010; Storni et al. 2015; Botero et al. 2020).
From the strictly analytical to the practical on to the speculative (Wilkie & Michael 2017), STS and design research traditions seem bound to cross paths, not least in relation to the techno-political endeavours for addressing societal challenges including food security, democratic participation, artificial intelligence, environmental concerns, health equity and so forth. We invite conference participants to take stock of the rich history of inspiration and dialogue between STS and design research towards expanding the joint horizons of disciplinary imagination and practice. The panel welcomes contributions that explore the refractions of design and STS sensibilities, including the uses to which STS terminologies have been (or can be) put into within design-related writing, research and creative practice, as well as the appropriation of design-based methodological approaches for STS fieldwork and thinking.
Converging topics and methods between STS and design could for example be tensions between efficiency and care, e.g. in health care or multispecies care; practices of wasting in relation to maintaining, repairing and composting; or topics that take stock of divergent methodological approaches, sensibilities and values, e.g. pertaining to use, culture or aesthetics in design.
By sharing current experiences and ideas, we hope this panel can contribute to ongoing and future cross-pollination between design and STS.

Signe Louise Yndigegn *Exploring careful interventions at the intersection of STS and participatory design*

This contribution to the panel “Taking stock: Design and STS sensibilities” explores the potential intersections between Participatory Design (PD) and Science and Technology Studies (STS), focusing on material culture, intervention and care. It delves into the issue of participation and engagement – and explores the idea of *careful interventions*.
In the presentation se considers how STS's critical theorizing of care can inform intervention-prone approaches within PD. By challenging method-

driven practice and questioning taken-for-granted assumptions, STS encourages a slower, more reflexive approach to research and intervention. This can provide the opportunity to create situational opportunities where care becomes a guiding principle that facilitates more nuanced and contextually grounded efforts.

Based on empirical material from PD events with a focus on material culture, intervention and care, we examine what careful interventions can be, as expressions of orchestrated efforts based on material engagements and care within design projects. We argue that given the inherent connection between design and care that often manifests itself in flawed and inattentive ways, delving and dwelling into this connection is a promising avenue that heralds new avenues for intervention and care engagement through design-oriented material practice.

Sissel Olander ***Reimagining STS as an experimental practice***

Over the last decades, the influence of STS on design research has been significant. This is especially true within contemporary experimental (co)-design research, where theoretical ideas and concepts from STS have provided a repertoire for practice-based design researchers inquiring into design processes, design methods, and design objects. STS has left its mark on the ways in which constructive design researchers understand and account for their work in empirical and analytical hindsight. The temporal orientation of design, in other words, designs preoccupation with what could be, and the radical experimental commitment that goes with it is however less explored. This presentation invites for a discussion of experimentation in STS, more specifically, the presentation inquires into the possible tensions between an empirical inclination and an experimental disposition. The presentation is based on a strategic design project in a maternity ward and revolves around the efforts of reviving a so-called in-house maternity clinic in collaboration with managers and midwifes. The exploration of what could be, in this case, the practical and collective work of imagining, conceptualizing, and forging an in-house maternity clinic, points to fundamental differences between the empirical and the experimental. This pertains not only to the orientation and interest of the researchers but also to the way research is set up, imagined, and done, and in turn, it circumscribes what counts as analytical. Thus, with this presentation I extend an invitation to STS scholars to explore a possible crossfertilization between STS and design in a reimagining of STS as an experimental practice.

Tau Lenskjold ***The question of culture in reassembling naturecultures after ANT***

Coming from a background in participatory and practice-based design research strongly influenced by ANT and feminist STS, I have for a good number of years been working in a research cluster centred on studies into the culture of design (Julier 2014) and design history. Setting up camp in this research environment has recurrently raised questions and discussions with my peers regarding attempts at thinking with, against or across the epistemological and ontological fault lines separating the study of cultures (encompassed by a priory items of analyze such as: ideas, attitudes, values, languages, practices, aesthetics, artifacts, institutions or structures of power) in conjunction with design, from the radical empiricist dictum "follow the actors" (Callon 1987) to study the effects of relations. To the extent the Culture has been addressed at all, it has been returned to the Social, another overarching signifying structure in dire need of deflation (Latour). Pointing to a "culture-blindness of ANT" Entwistle and Slater (2014, 162) argue that to 'reassemble culture' it is not enough to deflate culture to its materiel translations; we also need to pay heed to the multitude of practical ways in which human actors use conceptions of culture in making sense of, categorizing and performantly engage in relations with designed objects. Yet at the same time, the resistance of purification of culture by insisting on nature-cultures (Latour 1991) or naturecultures (Haraway 2003) recast the question to encompass what reassembling naturecultures might entail for the study of design.

Tim Cowlishaw ***Speculation through surplus: A Discard Studies approach for care through speculative design***

This contribution presents ongoing work which incorporates methods from the field of *Discard Studies* (Liboiron & Lepawsky 2022) into the practice of speculative design. Discard Studies has an explicitly speculative orientation - a normative commitment to "making new worlds" by "discarding well", as well as departing from a relational ontology of waste (seeking to examine broader sociotechnical "systems of waste and wasting") which has much in common with existing approaches in STS taken up in design practice.

In departing from the question "what must be discarded for this or that system to carry on?" *Discard Studies* provides tools for exploring *Counterfactual* speculations (Forlano & Halpern 2023). In addition its commitment to *situated* analysis of systems of power, its *normative* and *ethical* commitments, and its recognition of *ambiguity* and *incommensurability* in proposed solutions offer methods for situating

Speculative Design as a practice of care (Puig de la Bellacasa 2017, Pennington 2018), building on earlier work putting STS in dialogue with Latour's Matters of Concern (Lenskjold 2016). Taken together, these particular dispositions of Discard Studies offer a means of confronting the criticism that "critical and speculative design neglects the near and direct urgency of now" answering a "call to action to affect our collective present" (Ward 2021).

Through a comparison of the discard studies and speculative design literatures, illustrated by practical examples from my own work and workshops carried out with students on the BAU MA in Design Research, I will illustrate how methods from Discard Studies can be productively applied in speculative design practice.

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Panel 2

Waste, or the afterlives of repair? Temporality, materiality, politics
Chair: Proshant Chakraborty

This panel looks at the interrelationship between waste and repair through a temporal and material lens. It not only asks how waste manifests in a material sense from reparative interventions—broken objects, nonfunctioning components, organic matter—but also how waste can index the temporalities of other discarded projects, assemblages, and imaginations within largescale infrastructural networks (cf. Latour 1996). In doing so, this panel departs from the conventional anthropological idea of waste being "matter out of place" (Douglas [1966] 2002) and instead interrogates the presence, persistence, and politics of waste (Marder 2017). Following the recent scholarship on failure, repair studies, and discard studies (Carroll et al. 2017; Henke & Sims 2020; Martínez & Laviolette 2019; Lepawsky & Liboiron 2022; also, Liboiron 2021), it takes waste to be a productive analytic and ethnographic category, one that can address questions of morality, ethics, and responsibility. Paper presentations could potentially explore how material and temporal flows of waste speak to historical and contemporary political formations, such as in everyday practice and long-term policy planning. Following Lepawsky & Liboiron's (2022) theory of change, this panel's focus on the materialities and temporalities of waste and repair—its generation, management, elimination—invites further reflections on how political power is reproduced or challenged. The inquiry is animated by questions such as: In what ways are the materialities and temporalities of waste and repair mutually entangled? How do we make sense of waste generated in the pursuit of socially desirable goals, like efficient and sustainable technologies? Conversely, how can we use waste to understand the discarding of some sociotechnical alternatives in favour of others? Potential paper submissions could deal with waste alongside the following themes: capitalism and coloniality, care ethics and emotional labour, heritage and sustainability, in/formal and il/legal practices, infrastructural (non)development, social and ecological sustainability, state and transnational actors, to name a few.

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Trine Brox ***Violent smashing, careful repair: Waste imaginaries and trajectories in the Cultural Revolution and its aftermath***

This presentation will highlight how waste is a product of not only material processes, but also other forces (e.g., political, moral, religious) that move people to take specific action – whether it is violently smashing a thing or carefully repairing it. I will discuss smashing and repairing as political and moral acts, ways of dealing with problematic things (complete or broken). My example will take us back to mid-twentieth century Lhasa and conceptions of waste at an extraordinary time of economic and symbolic iconoclasm. I show how the violent destruction of Buddhist material culture during this period involved sorting practices through which various actors distinguished between the excessive, worthless, valuable, and so forth. A Tibetan entrepreneur's account of how he repaired Buddhist statues from rescued parts of smashed statues in the 1980s draws our attention to the durable value of broken things when they are considered sacred or efficacious. Such waste can be considered worthy of becoming what I call "waste as afterlife" (Brox 2022), having productive "sacred" afterlives when repaired and restored in specific ways. Whether from the perspective of Communist ideology or Buddhist beliefs, the power of these objects to demand such purposeful action (violent destruction and respectful care) presents a powerful prism on waste as a moral-political process.

Samuel Moses Ntapanta *Repair Allegories: Phenomenology of Electronics Repairing in Tanzania*

In Tanzania, objects do not go abruptly from brokenness to waste; instead, they travel back and forth between repair regimes, creating specific ways of being and engaging with the world. Repairing is an everyday experience for many people in Tanzania. A repairer is always available to tinker with anything. However, acts of repair or finding a repairer do not only appear when things break down. Repair is considered a part of an object's life and people´s life with an object. As a result, people chose certain products,

relying on particular infrastructures and the availability of repair before breakdowns happen. These possibilities and the ability to repair are considered and searched way before acquiring the object. This contrasts with a radical emphasis in other economies between the whole, intact and functioning object, and the one that is broken, apart, and not working – in need of repair or wasted - needed out of sight. In Tanzania, for most people, objects are never really whole and never really broken; even when brokenness happens or become obsolete, objects retain the potentialities of a life and the intimate relations that the object gathers with the people. And there is a way to restore those relationships. In this paper, I want to tell stories of repairing, of people and of objects. These gatherings (of repairing, electronics and people) that happen in disparate places are central to living in a planned obsoleted world.

Julie Nygard Solvang ***Displaced responsibility: the transportation and transformation of discarded electronics and electronic waste***

Discarded electronics move. When fridges, computers, and other electrical products are thrown away, they do not stand still. The movement can involve the crossing of several national and legal borders, as well as intricate social dynamics that re-arrange, dismantle, and produce novel perceptions of these materialities. In these trembling moments and movements, where breakage is temporary, the entanglement of waste and value becomes clear. While the transportation and transformation of electronics enable repair, re-valuing, and re-use of objects and materialities, it simultaneously displaces responsibility for the environmental costs of electronic waste (e-waste) in spatial and temporal terms. This paper explores social dynamics of transportation and transformation, as it follows the itinerary of discarded electronics and e-waste from warehouses in the outskirts of Hamburg, Germany, to the sidewalks and dumpsites of Accra, Ghana. Based on ethnographic fieldwork conducted with people that move such electronics along the way, it backtracks this illicit trade flow in the grey zones of global post-colonial circuits.

Anna Bohlin and Veera Kinnunen ***Using, keeping, wasting: Negotiating material temporalities in the more-than-human household***

This paper focuses on the role of domestic activities associated with waste generation and prevention, widely described as being of central importance for the transition to more sustainable consumption practices. Taking a more-than-human view of the household, a key site for understanding what, how and when stuff becomes rendered as waste, it argues that temporality is a central but overlooked aspect in the scholarship on domestic consumption and discarding. Following Mol, who calls for a

situated, more-than-human politics of labour, “a matter of ongoing, practical, simultaneously social and material negotiations” (2021: 128), we explore the coordination (Gan & Tsing 2018) of the temporal rhythms, cycles, sequences and speeds that characterise the alignment and transformative entanglement of humans and nonhumans in the home. Households can be seen as meshworks of temporal lifelines: our own, but also of a range of other living and non-living entities and organisms (Ingold 2011). Drawing on ethnographic fieldwork in Sweden and Finland, we argue that patterns and practices of consumption and waste creation are closely related to our ability to be attuned to such temporalities (cf. Gregson 2007). Perceiving, sensing and imagining signs of transformation and anticipated change, we adjust to the temporalities of objects and organisms, assessing their lifespans and windows of usefulness and value, and in turn, negotiate and make use of these to deal with our own becoming and changing. Given that it can both drive and slow down consumption and waste generation, understanding such negotiation of domestic material temporalities is key to finding sustainable ways of using, keeping and wasting.

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Pankaj Sekhsaria ***Waste in/as reconfigured materiality: The story of a cutting edge scientific instrument and a jury-rigged rural automobile in India***

This story is the biography of two technological objects made in two different parts of India. One is a cutting-edge instrument made in a modern scientific laboratory; the other a set of jury-rigged visible-invisible automobiles made locally and informally in a distinctly remote setting.

A granular and detailed study of these objects – how they came about, who makes and uses them and what networks are mobilised to bring them to life – reveals striking connections and commonalities despite their apparently wide differences. They operate largely in informal spaces, constantly configure and reconfigure materialities of objects discarded as wasted and are seen responding intimately to local contexts and resource constraints.

This, even as they simultaneously service and exploit the vast spaces left empty by formal institutional structures and mechanisms.

This happens, the paper submits, because at play here are a rich set of democratic practices of technological innovation, which in turn are linked to and mobilised by a ‘culture of innovation’ that is deeply contextual. Both however remain invisible or are deliberately made invisible because they don’t align with certain imaginations and imaginaries of innovations and innovators and of what it means to be technological advanced.

Panel 1

Mapping and valuing excess in the food system (workshop)
Chair: Stine Rosenlund Hansen

There seem to be uncountable cracks in modern food systems, where resources are leaking. A never-ending pursuit of efficiency and scale, and a linear supply chain thinking has led to excess and waste of food in some countries, whereas food loss in broken value chains is widespread elsewhere. The later years has seen increasing discussion on the use of side streams and bio resources, thereby introducing food waste into valuation-processes. Paradoxically, finding solutions for using waste risks to create a pull for increasing excess food production.

Thinking with the concept of excess in relation to food systems can follow other trails. For example, nutrients are produced and applied in excess in many farming practices with water pollution as a consequence, while the composition of nutrients within many food items is unbalanced with excess of some and lack of others. In another vein, many countries have gotten use to constant and easy availability of the food that fit all preferences (excess of choices?), with overproduction and high food-miles as some of the consequences. Simultaneously, the resources going into food production is not currently mirrored in how it is valued (monetary, culturally, etc), with low inclination to care about the (wasting of) food.

Contributions to this panel can fall within the following themes:

- Theorization and analysis of paradigms leading to waste or excess in food systems, including degrowth and posthuman studies
- Approaches to understand different versions of excess and waste in food systems, including nutrients in farming and eating practices
- Socio-material analysis of human-food relations leading to waste, including enactments of food quality, access, and availability.
- Methodological approaches to explore problems of excess in the food system, especially within mapping approaches.
- Valuation of food and food waste, including dilemmas arising from practices of valuing food waste, such as biofuel, or different aspects of food valuation processes, such as financial, handling, naturalness, historical etc. leading to waste, loss and/or excess.

The panel will be held in a workshop format with short presentations and shared discussions, potentially in break-out groups.

Tanja Schneider ***Financing FoodTech Futures: The assetization of food innovation***

In recent years, entrepreneurs in Silicon Valley and beyond have been developing a range of novel foods – so-called FoodTech innovation – with

the aim to transform the production and consumption of food. Examples of recent FoodTech innovation include plant-based milks and meat alternatives, cellular meats, produce grown in vertical farms and insect-based foods. In this talk I explore how entrepreneurs make FoodTech innovation valuable for venture capital investors who financially support FoodTech start-ups. I trace the financial logic guiding these high stakes investments, examining how it shapes which innovations receive funding and which food futures are realised. My analysis builds on previous scholarship attending to assetization (e.g., Birch 2019, Muniesa et al. 2017), valuation studies (e.g., Dussauge et al. 2015; Kornberger et al., 2015) and the political economy of novel foods (e.g., Wurgaft 2020, Sexton 2020, Fairbairn et al. 2022). I argue that an 'assetization' (Birch and Muniesa 2020) of FoodTech innovation is taking place: that food innovation is being turned into an asset which can be controlled, traded and capitalized as a (future) revenue stream. Using fieldwork data, I illustrate how FoodTech innovations are made valuable through sustainability framings. I also reflect on what is meant by sustainability in relation to FoodTech innovations, drawing on field notes, interviews and documentary evidence collected for my 'valuography' (Dussauge et al. 2015) of FoodTech innovation, to be published as the monograph Venture Food. Through this, I show how the financial logic guiding FoodTech investments and the assetization of novel food require the continuous practical accomplishment of FoodTech's valuation as a 'sustainable innovation'.

Aline Stehrenberger ***Constitution of Databases: The (Un)Predictability of Food Waste***

Apps that aim to reduce food waste by redistributing it are an increasingly popular way of countering excess food. In this paper, I explore the case of an app in the making that focuses on altering grocery shopping behaviour to reduce household food waste. I ask how the constitution of its database reconfigures practices of food waste reduction. 'Ground truthing', defined as the constitution of databases (Jaton, 2017), helps me to frame the development team's valuation processes around their data gathering. After the literature review, I introduce my case, WasteMentor1, a research-based app with the goal of conducting digital receipt-based sustainability and food waste monitoring and interventions. This app builds on grocery shopping data collected through customer loyalty programs of two major Swiss retailers as well as a 'food waste collection study'. I have taken an in-depth ethnographical approach to studying the development of the app's infrastructure (Star, 1999; Hine, 2015; Schneider & Eli, 2021). Analysing how the discarded food is turned into data and combined with digital

receipts to eventually reach the ground truth, I argue that the narrow focus on mainstream retailers neglects how food (waste) is enacted in practice. I conclude with a critical reflection on the attempt to datafy and predict (potentially) decaying food, whereby I aim to contribute to critical infrastructure studies.

Molly Donaldson ***One Copenhagener's food waste is another's treasure" - An exploration into the circular foodscape of Copenhagen***

Renowned for its sustainability credentials, Copenhagen is often cited as one of the greenest cities in the world. Despite this, the Danish capital continues to face the pressing problem of food waste, as food outlets drive the overproduction of food. In response, community and market-based initiatives have emerged, redistributing, sharing and transforming food surplus to prevent these valuable resources from going to waste. This paper explores the extent to which four food rescue organisations tackle food waste produced at the municipal level through the enactment of circular food practices. In the second portion, the limitations of the community and market-based initiatives are addressed, followed by the potential tensions of upscaling food rescue operations in the city. To answer the research questions one-to-one interviews are used, in combination with a spatial analysis of the circular foodscape of Copenhagen. The results reveal how particular community and market-based initiatives significantly reduce food waste, inciting multiple actors to change their linear food relationalities by adopting less wasteful, more circular food practices. In turn, this exploration into the murkier realm of food waste unearths a circular foodscape driven by food rescue organisations, as their circular food practices feed into the growth of the circular food economy of Copenhagen.

Anne Clausen ***The relationship between food waste and quality requirements in public food procurement***

With the Sustainable Development Goals (SDG) adopted in 2015, the EU countries are committed to meeting the SDG 12.3 targeting to halve per capita food waste by 2030 and reduce food losses along the food production and supply chains. I present results from a 4-year long cross-sectoral research project which focus to understand the relationship between amounts of food waste generated as an unintended consequence of the public food procurement process and the related legislative requirements of contract. As well as the relationship between food waste and the everyday practices across the food supply chain. Within the theoretical frameworks of Practice Theory, analysis reveals that high expectations, as well as legislative requirements on food quality causes

large amounts of food waste, especially in the food categories of ‘dairy’ and ‘fresh fruits and vegetables’. Strict requirements on shelf live challenges dairy products leading to milk being wasted with 3+ days left in shelf life. While requirements on cosmetic specifications, year-round availability, broad selections of single produce and requirements on stockkeeping units of all items on the quotation list challenges the waste of fresh fruit and vegetables. Through cross-sectoral Living Lab approaches, the project has investigated and evaluated practice-near possibilities of reducing food waste instead of pushing the problem further down the public food supply chain. The project offers unique insights into the complex multitude of everyday practices related to the paradoxical connection between legislative requirements on high food quality and the waste of tons of usable foods across the private-public food supply chain in Denmark.

Open panel B

Waste and value through another lens
Chair: Amanda Obitz Mogensen

Julie Kordovsky and Morten Nissen ***Redrawing watelands of elderly care***

'Human capital' works too well as a concept with which to describe policies of 'competition states'. To this concept inheres not only a specific focus on valued attributes, but also specific mechanisms of misrecognition, of overlooking or disregarding what is no longer useful, not entirely unlike processes of garbage disposal and the cultural disdain for those who take care of it. Although 20xplic would subscribe to the cynicism of regarding elderly care as waste management, it is hard to dismiss the hypothesis that the economic logic, which gives the expansion of labor force top priority, often with reference to growing numbers of old people, is the same that allows very limited resources to municipal elderly care.
Against these forces, how can a future wider care for life, including human life in its entirety, be enacted by constructing caring regards? In this paper, we seek to contribute to this by taking literally these visual metaphors. We ask: Could the wastelands of elderly care begin to be redrawn (cf. Ingold) in a caring way by including drawing in the education of potential care workers?

The idea may look weird for anyone who has not engaged with aesthetic theories that suggest art as ways of reconfiguring sense and in-/capacities (Rancière, Adorno) or of constructing the social (Groys, Bourriaud). Recently, Morten Nissen has analyzed a poetry 'writing group' as post-therapeutic, as a way of moving addiction treatment from optimized cure to a wider care for cultivating motives (in Rearticulating Motives). In this presentation, Julie Kordovsky will describe shared knowings and visual works from her fieldwork with trainees in elderly care, where doing drawing is introduced Into the learning ecologies of the practical training in nursing homes. In a joint research creation (Manning) namedThe things they live with, Julie Kordovsky and a group of trainees assume a relational epistemology (Cozza) in their investigations of the materiality of elderly care.
Together, we – Julie Kordovsky and Morten Nissen - will point towards what we envision to be the potentials and perils in possible future attempts at redrawing the wastelands of elderly care.

Joren Peeters ***Beyond the Catchphrase: Matter and meaning, or, why we need to make waste dirty***

In her seminal *Purity and Danger,* one of the foundational texts of discard- and waste studies, Mary Douglas famously defined dirt in a catchphrase: "dirt is matter out of place". According to Douglas, something is dirt, not because of some natural feature of the object itself, but because the object doesn't fit the way a culture or collective tries to schematize the world in a symbolic order. Inquiries into the problem of waste must relate themselves to Douglas' fundamental insights, be it as point of departure or point of critique. As point of departure, Douglasian dirt is used as a general frame to analyse different waste problems. Here, dirt and waste are equated. As point of critique, the specificity of Douglasian dirt is stressed. Here, dirt is being understood as something different than waste and often seen as irrelevant to tackle the problems of our contemporary "discard society": Douglas' focus on symbolism fits awkwardly with the billions of tons of waste that arrive daily at the world's landfills.

In this paper, I opt for a middle way: I argue that it is precisely *because* of its specificity that Douglasian dirt is a useful category for analysing contemporary waste problems. More precise: by distinguishing dirt from waste, I want to stress the critical potential of dirt. I argue that dirt and waste are located on different ontological levels: dirt is about meaning and waste about matter. To articulate this difference, I use the distinction the French sociologist Luc Boltanski makes between "world" and "reality". "World" is the concrete and confuse *flux of life* and "reality" is the fragile *semantic order* in which we try to pinpoint the world to enable communication and meaningful interaction.

By closely looking at Douglas analysis and the examples she gives, we see that the order she describes (and which defines dirt by excluding it) is *reality* in Boltanski's sense; it is situated on the semantic level and only has a material and spatial significance to the extent that matter and space are experienced and organised *as meaningful*. Thus, contrary to what the catchphrase suggests, dirt isn't in the first place about matter, but about meaning: dirt isn't "matter out of place", but "anomaly", that which doesn't fit the fragile *semantic order* in which we try to pinpoint the world as meaningful. This means that waste, the concrete material discarded by the production processes polluting our planet, is not necessarily dirt. Furthermore, a central problem of our time, is that *most* waste is not dirt. Most waste is in the *reality* dominating our contemporary *world* not an anomaly but a calculated outcome of production. I will conclude by arguing that if we want to alter dominant systems of waste and wasting, we'll have to make specific waste materials dirty. This means that, instead of managing certain waste materials and making them meaningful through pseudo-sustainable measures such as recycling, we'll have to imagine new realities in which there simply is no meaningful place for certain specific waste materials.

Savannah Schaufler, Tatiana Konrad, Harald Wilfing *Plast(dem)ic': Bridging Awareness and Action*

Plastics are praised for their versatility yet criticized for their destructive impact on the environment. This paper explores the persistence of plastic waste, driven by industrial influences and societal convenience, that ultimately infiltrates our ecosystems and bodies. Introducing the term "plast(dem)ic," this paper highlights the urgent and widespread challenges associated with plastic pollution that parallel the rapid spread of a pandemic. Similar to a pandemic, plastic pollution affects communities globally, visibly impacting environmental health and exacerbating social and environmental inequalities. Employing posthumanist, materialist, ecological, and behavioral perspectives, this paper examines wasteful behaviors within cultures of disposability. Through the lens of the cognitive dissonance theory, it examines the gap between awareness and action regarding wasting behaviors. Advocating for interdisciplinary approaches, we propose integrating behavioral insights into environmental education to facilitate a transition to more sustainable practices and solutions to "plast(dem)ic." By emphasizing the importance of interdisciplinarity, this paper provides a detailed analysis of waste dynamics in contemporary societies characterized by disposability.

Thomas Roberts *Recreational litter picking won't save the world, but….*

Traditionally clearing up waste has been seen as 'dirty work', something which people only engage in when they have few alternative options. However, in recent years we have seen a huge growth in recreational litter collection, where individuals and groups commit significant amounts of their spare time to clearing up other peoples waste. This waste collection takes a variety of formats and is often combined with a range of other activities such as running (plogging), scuba diving (removing ghost fishing gear) canoeing/paddle boarding (removing rubbish from inland waterways and coastal areas). While recreational litter collection is unlikely to make a significant impact on the global waste problem, it can make a noticeable difference to local areas. Volunteer litter pickers are also having a significant impact raising awareness of waste related issues at both a local and global scale by sharing pictures and films of there activities on social media. Furthermore, these posts are increasingly being picked up by the mainstream media and encouraging wider discussions about the problems caused by waste. In addition to the impact recreational waste collection is having on the waste problem, this research has found significant benefits to the individuals involved related to both mental and physical health. In this

paper I will present the findings from a recent ethnographic study into recreational litter picking and explore the role it could play in tackling the contemporary waste problem.

Panel 3 session 2

Taking stock: Design and STS sensibilities
Chair: Tau Lenskjold

For more than a decade, diverse areas of design research have been taking a concerted interest in STS as a source of theoretical and methodological inspiration. STS, it can be argued, has always been concerned with design practices and processes, through which the social, political and technological are imagined, constructed and realised (Danholt & Halse 2016). This has led to investigations into specific design practices and domains, such as the design studio (Farias & Wilkie 2018) and user representations (Hyysalo, Jensen & Oudshoorn 2016), and to appropriation of design-based methods, such as the design prototype (Jiménez 2014). STS-based theoretical frameworks have in turn been put into use as analytical strategies across design research, from design history (Fallan 2010) and design culture (Kimbell 2012) to participatory design (Björgvinsson, Ehn & Hillgren 2010; Storni et al. 2015; Botero et al. 2020).
From the strictly analytical to the practical on to the speculative (Wilkie & Michael 2017), STS and design research traditions seem bound to cross paths, not least in relation to the techno-political endeavours for addressing societal challenges including food security, democratic participation, artificial intelligence, environmental concerns, health equity and so forth. We invite conference participants to take stock of the rich history of inspiration and dialogue between STS and design research towards expanding the joint horizons of disciplinary imagination and practice. The panel welcomes contributions that explore the refractions of design and STS sensibilities, including the uses to which STS terminologies have been (or can be) put into within design-related writing, research and creative practice, as well as the appropriation of design-based methodological approaches for STS fieldwork and thinking.
Converging topics and methods between STS and design could for example be tensions between efficiency and care, e.g. in health care or multispecies care; practices of wasting in relation to maintaining, repairing and composting; or topics that take stock of divergent methodological approaches, sensibilities and values, e.g. pertaining to use, culture or aesthetics in design.
By sharing current experiences and ideas, we hope this panel can contribute to ongoing and future cross-pollination between design and STS.

Harun Kaygan ***Robots, trains, playgrounds: Mapping user bodies in design and technology***

Design research and STS have routinely intersected in their theoretical interest in the ethnographically specific descriptions of the ways in which people (are made to) interact with designed things, from highway underpasses to telecare technologies. A relatively underexplored intersection has been that of the user's body as it is addressed by design practices and objects (notwithstanding references e.g. to Mol's multiple bodies, Haraway's figurations, and more recently to the STS work in care

studies in design research). With the objective to address that gap, the presentation derives from a range of design case studies, including doctor-designer collaborations in medical product development, gendered experiences in public transportation, and representations of the child in playground design, in order to map out some of the intersections that became apparent in this work. The objective is to show that there is ample opportunity for dialogue between the two disciplines in their discussion of the status of the body in relation to the societal role and impact of design and technology development. I suggest we rethink sociology of user representations in embodied terms, triggered by the heightened awareness of and methodical engagement in design and development activities with embodied creativity, interaction and collaboration on one hand, and continue being inspired by the work in care studies and feminist research in STS on the other. Doing that, I also extend an invitation to STS scholars to demonstrate persistent critical interest in designed interaction, including its embodied, aesthetic qualities.

Li Jönsson, Åsa Ståhl, Kristina Lindström *Grief, hope and holding surplus house*

This contribution to the panel "Taking stock: Design and STS sensibilities" activates socio-materialities that are haunting our contemporary and future livelihoods. The authors approach the remnants and potentialities of the era that has been marked by progress with three main concepts: grief, hope and surplus.

The authors will share key moments from two recent and ongoing design research projects *Grief and Hope in Transition* and *Holding Surplus House* and take them as starting points for how design could engage in unlearning its own history through reading history differently. This is done with the help of a combination of artistic research methods and science and technology studies that are currently developed in the ongoing research environment *Design after Progress*: *Reimagining Design Histories and Futures*.

Merle Ibach *Shredding my PhD research on DIY communities: knowledge production and waste in design research*

During my PhD research on DIY communities, exploring how they prototype degrowth futures, I produced not only a dissertation but also a lot of words scribbled and printed on a lot of paper. Printing out my drafts to work my way through my thoughts, reiterating them again and again, until a somewhat consistent argument was built, was a necessary, thus paper-heavy process. The collected leftovers of my knowledge production

witnessed a draining and energy-consuming process, both for me and the environment. By shredding my research notes, and making paper from it again, I reviewed my entire argument. This article reflects on the physical and emotional labor involved in crafting arguments and the subsequent disposal in the production of knowledge. Waste is a main arena for critique in design, and has been since the early days of industrial production (van Helvert 2016), i.e. the Arts and Crafts movement criticized mass consumption by returning to crafts, whereas the capitalists criticized Morris‘ handmade paper and books as „conspicuous waste“ (Veblen 1899), requiring expensive labor, and preventing the access and democratization of knowledge. The notion of waste is strongly linked to modernist, capitalist imaginaries of production, commodity, wealth, efficiency, and class (Miller 2016). Today, in the aftermath of an industrial era, when confronted with the ruins of capitalism and the pollution of the world we live in (Lindström/Stål), waste becomes again an arena for design criticism, but with different signs: the massive amounts of waste are no toxic byproduct of our wealth anymore, but the mere fabric of our existence (Midal 2019). Drawing on STS sensibilities, particularly the notions of knowledge production and care (Mol et al. 2010, Puig de la Bellacasa 2017), ripping, shredding, mashing, watering, blending, and destroying my carefully crafted notes for good, unveils a physical and emotional exhausting endeavor in reframing waste to unmake capitalist imaginaries in the production of
knowledge.

Morten Krogh Petersen & Victoria Ankerstjerne *Designing with temporalities: The case of Diaper System 2030*

We have just presented the setup of the multidisciplinary and cross-sectoral research and innovation project, Diaper System 2030, to an interested group of public sector innovators. “Why do you have ‘2030’ in the title of your project?” one puzzled listener asks. We look at our project partner from a waste management facility. He answers that the project ends in 2027 – after that, we have three years to get the future system up and running. The answer satisfies. Over the coming three years, Diaper System 2030 will develop and test a fully circular diaper system across sites of commercial diaper production, elderly care, and waste management. With the timestamp in its title, the project performs a specific version of the future well-versed in design research: the future as an outcome of, for instance, ‘designerly’ making (Ehn et al. 2014), rehearsing (Halse 2010), or, simply, futuring (Fry 2008). A growing body of STS literature, however, explores “the constructed nature of time to point at the multitude of temporalities” (Marquardt & Delina 2021: 3), urging us to explore the

project's 'designerly time' as one specific temporality that might collaborate or clash other project-related temporalities. We give accounts of such temporalities worked up through our initial empirical engagements with the practices the project seeks to understand, connect, and transform. We use these examples to explore the sentiment that our contribution as futuring design researchers might end up as waste if not coupled with STS-informed understandings of the enactment and coordination of the temporalities of design, diaper production, care, and waste.

Panel 2 session 2

Waste, or the afterlives of repair? Temporality, materiality, politics
Chair: Proshant Chakraborty

This panel looks at the interrelationship between waste and repair through a temporal and material lens. It not only asks how waste manifests in a material sense from reparative interventions—broken objects, nonfunctioning components, organic matter—but also how waste can index the temporalities of other discarded projects, assemblages, and imaginations within largescale infrastructural networks (cf. Latour 1996). In doing so, this panel departs from the conventional anthropological idea of waste being "matter out of place" (Douglas [1966] 2002) and instead interrogates the presence, persistence, and politics of waste (Marder 2017). Following the recent scholarship on failure, repair studies, and discard studies (Carroll et al. 2017; Henke & Sims 2020; Martínez & Laviolette 2019; Lepawsky & Liboiron 2022; also, Liboiron 2021), it takes waste to be a productive analytic and ethnographic category, one that can address questions of morality, ethics, and responsibility. Paper presentations could potentially explore how material and temporal flows of waste speak to historical and contemporary political formations, such as in everyday practice and long-term policy planning. Following Lepawsky & Liboiron's (2022) theory of change, this panel's focus on the materialities and temporalities of waste and repair—its generation, management, elimination—invites further reflections on how political power is reproduced or challenged. The inquiry is animated by questions such as: In what ways are the materialities and temporalities of waste and repair mutually entangled? How do we make sense of waste generated in the pursuit of socially desirable goals, like efficient and sustainable technologies? Conversely, how can we use waste to understand the discarding of some sociotechnical alternatives in favour of others? Potential paper submissions could deal with waste alongside the following themes: capitalism and coloniality, care ethics and emotional labour, heritage and sustainability, in/formal and il/legal practices, infrastructural (non)development, social and ecological sustainability, state and transnational actors, to name a few.

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Marisa Cohn ***Software out of time: temporal regimes of evolvability***

In software development, "waste" primarily refers to *wasted effort*, or the entropic forces that disrupt productivity. Rubrics point to the sociotechnical causes of software waste such as building features of no value, burdening teams with unwarranted scope or complexity, "thrashing" changes caused by churn of features or people, and loss of knowledge or time due to waiting, hand-offs, task switching, or loss of team members (Sedano et al 2017). Waste is also caused by what developers refer to as software cruft, code rot or stink, and architectural decay, that arise as software accrues over time. "Sustainable software" (defined as sustained growth with limited disruption) is thus synonymous with eliminating *waste*. However, this is challenged by both the ease of growing new code, and the difficulty of discarding older legacy software components. Software's durability creates troubles for ongoing disciplinary work to not only maintain systems but recursively maintain their maintainability and ensure their continuous "evolvability". Evolvability requires temporal bridging between the speculative work of deferring repairs with the need to maintain backwards compatibility to parts of the system one cannot yet discard. Software maintenance is thus filled with contestations over different forms of reproductive labor and care work – over what to allow to obsolesce into disrepair, what to actively cut, and what to continue to support. Naming what is wasteful involves negotiating the sense that some lines of code are not so much "out of place" but "out of time." Attention to waste reveals how *evolvability* as a disciplinary regime entail distinguishing virtuous repair from pathologized attachments to obsolescent and material labor by associating some care work with decay, rot, and gendered tropes of poor aging.

Laura Kocksch and Mette Simonson Abildgaard ***Wasting Time as Maintenance – Striving for Fragility in Breaking Digital Infrastructures***

The technical crew had been stuck on site for five weeks. Sitting next to several containers filled with everything they need to establish a new

satellite ground station, they waste their time watching daytime television. They are waiting, biding their time until the carpenter arrives who will isolate the building they are installing the equipment in.

"Maybe … we do nothing", the IT manager of an industrial printing company in Eastern Jutland sighs. He had explained how his company struggles to improve the cybersecurity of their aging and outdated truck-size industrial printers, and now shily wrangles his hands. The printing business has been declining for years, what is there to secure?

In this paper we delve into practices of maintaining digital systems that are not aimed at readiness, improvement or progress, but uneasily and painfully to our protagonists, revolve around waiting, inactivity or hesitation. Our ethnographic field sites encompass cybersecurity practices in small- and medium-sized enterprises in Denmark and telecommunication and bandwidth connectivity in Northern Greenland, both brimming with 'ghosts' and 'monsters' (Tsing et al. 2017) of past and conflicting digital infrastructure logics. In times of alarmism, perpetual crisis and innovation, doing nothing is hard and moralized. Leaving things be requires much. Not striving to get ready, grow or move on does not mean doing nothing. It is not careless or malicious, nor easy. Digital infrastructures emerge painfully fragile from this view, slightly better or worse, but without light at the horizon.

Daniel Nordstrand Frantzen and Julia Kirch Kirkegaard *Revaluing wind energy by putting its waste to work – the case of Power-to-X*

Following calls to treat waste as an analytical category, which can shed light on how entities become treated as disposable or indispensable (Liboiron & Lepawsky, 2022), we use 'waste' as a lens to explore both the societal valuations and criticisms of wind energy as well as recent efforts at maintaining the value of wind energy through Power-to-X (P2X) technologies. To better grasp the relationship between energy and waste, we turn to Daggett's *The Birth of Energy* (2019), which delves into the moral underpinnings of energy consumption, where using energy to put something to work is justified while wasting energy is considered bad. We view this work-waste-dichotomy as an important 'register of valuing' (Heuts & Mol, 2013) wind energy. We argue that wind energy has become highly valued in society due to its cost-effectiveness, renewable nature, and minimal waste compared to fossil fuels. However, we trace a growing critique that energy goes to waste due to the inflexibility of wind energy: Electricity must be used instantaneously since it is difficult to store. We see P2X as an attempt to recuperate (Chiapello, 2013) this critique by altering the temporalities of wind energy, allowing it to be stored in the form of fuels like hydrogen that can be put to work in the future. However, P2X

brings new controversies of waste, both because the conversion efficiency from wind energy is poor and due to concerns that hydrogen production may still rely on fossil fuels despite being marketed as ‘green’.

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Julie Funk *Xenohormones: The Colonial Condition of Living in/as Chemical Excess*

In the scientific paradigm that situates hormones as “chemical messengers,” this class of chemicals is understood as a type of bioinformation capable of mediating human life. When undesirable messages or noise, such as endocrine-disrupting chemical waste and other xenohormonal toxins, transmit through the human body, these chemicals are considered “signal jammers,” harming reproductive health and fertility (Colborn et al., 1996). Such “xenological” forms of mediation are often toxic and disproportionately affect racialized communities (James-Todd et al., 2016). In this presentation, I explore what it means to be burdened by xenohormonal toxicity under colonial flows of capital that produce endocrine-disrupting chemical waste and direct them towards racialized life, such as the petrochemical industry surrounding Aamjiwnaang First Nation in Canada and in nearby Detroit in the United States. Such orientations to chemical waste not only assume the land they pollute is surplus and as an appropriate container to direct the endocrine-disrupting excesses of oil refining and manufacturing into (Liboiron, 2021) but also treat those who occupy the land similarly as “living forms of waste available for destruction” (Murphy, 2017) through the mediation of toxic xenohormones that threaten reproductive futurity. From this colonial assumption of the excessive and disposable racialized body under the toxic burden of xenohormones, I turn to the cosmogeny of Black, Detroit-based electro music group Drexciya as an aquatic site in which the xenological

hormonal and reproductive politics of colonial capitalism during The Middle Passage are reinterpreted and redirected outside of excesses of population or pollution.

Panel 4

Getting wasted with generative AI (Workshop)
Chair: Anders Kristian Munk

The large language models powering the current generation of generative artificial intelligence have become intricately connected to waste. Not only do they stand accused of all manner of wastefulness ranging from the energy spent training them to their potential for wasting our attention, they are also born out of an abundant approach to AI where more, not less, has indeed, to the surprise of many, proven to be more. At least for a while, in early 2023, meticulously curating high quality training data appeared to be a waste of time and a thing of the past as generative pre-trained transformers, built on very large corpora of unknown provenance, took the world by storm and demonstrated almost magical abilities. This is now itself a source of controversy as the need for better curated and less wasty data practices becomes evident (less is to some extent more again), as focus is put on the often invisible labour required to repair and maintain the models, and as the idea that training data found 'lying around' on the internet is just a way of putting other people's data waste to good use become challenged by copyright claims. On top, having a conversation with a generative AI is arguably a wasted experience in its own right, in the sense of being inebriated, rambling, and hallucinating. In this panel we invite experiments and reflections on what we can learn from hanging out with large language models. Sometimes, getting wasted together reveals deeper truths about ourselves and the way we see each other. What is at stake when we look beyond generative AI as wasteful? Beyond garbage in/garbage out? What if we take the models seriously as companions in our research practices? Who are they? How can we get to know them? And what kinds of relationships should we expect (and demand)?

Johan Irving Søltoft & Anders Koed Madsen ***Experimenting with mundane synthetic politicians***

In urban settings, connecting with local political agendas can be challenging. Existing methods for interfacing with politicians are often mediated by journalists or involve staged events like rallies or town hall meetings, which may not reflect everyday mundane politics. The Copenhagen Municipality offers online access to its agendas and minutes, yet the vast volume makes meaningful engagement with this material difficult for citizens, resulting in the data being underutilized or wasted despite its potential value.
To address this, we want to facilitate a live making and doing experiment utilizing a generative AI model to facilitate conversations with synthetic mundane politicians. Researching the unique way political communication occurs with generative AI, using this Q&A format, to

investigate how it distinguishes itself from existing modes of political discourse between citizens and politicians. Can generative AI enable a more informal dialogue between the public and politicians, like interpersonal interactions, rather than the traditional rhetoric used in media or town hall meetings? What happens if we take these synthetic politicians seriously? What can we learn from spending time with them?

Tim Cowlishaw ***Neural-network necromancy: An experimental media archaeology of language models***

Large Language Models feed on waste, in terms of the abundance of data used to train them, an act of recycling, or a transfer of value (Thompson 2017). But this isn't the full extent of LLMs relationships with waste - taking a view informed by *Discard Studies* and asking "what must be discarded for this or that system to carry on?" (Liboiron & Lepawsky 2022) offers the possibility of problematising their seemingly unstoppable advance, and asking, in the popular refrain of STS, how things might *"be otherwise"* (Woolgar 2014).

One stream of waste generated by LLMs are discarded 'obsolete' models, regularly superseded. Advances in generative AI are often couched in technological determinist terms - a race towards an inevitable 'artificial general intelligence'. Older iterations, once touted as the state of the art, are quickly discarded and forgotten.
Paying attention to these discarded ancestors of the current 'state of the art', following Siegfried Zielinski (2008), challenges this idea of inexorable progress and the apparent "necessity for simple technical artefacts to develop into complex technological systems".

In this activity, building on earlier work carried out at the CCCB in Barcelona (Cowlishaw 2024), we will investigate these alternative trajectories through an Experimental Media Archaeology activity (Fickers 2013). Participants will interact with 'obsolete' language models, from markov chains to GPT-2, training, fine-tuning, and conversing with them, probing their affordances and relations, and generating *counterfactual histories* (Forlano 2023) of AI to speculate about alternative, less extractive, AI futures.

Anders K Munk, Mathieu Jacomy & Matilde Ficozzi ***How to order a drink in GPT***

Etiquette books have always been a key source of interest for various cultural studies. A prominent example is Norbert Elias' history of manners.

Similarly, manners have been keenly studied ‘in the wild’. In the 1960s, for example, the ethnoscientist tradition in anthropology framed the study of culture as the search for underlying behavioral algorithms that would allow you to fit in to a given social group. A well known example is Charles Frake’s 1964 paper on how to ask for a drink in Subanun.

In this workshop we will collectively and comparatively explore how to ask for a drink in GPT 3.5, Gemini, Mistral, and other popular large language models in an effort to discover their cultural preferences. The objective will be to prototype a series of etiquette cards on how to behave like an LLM at the conference dinner. We hope this will facilitate a conversation about what it means to understand LLMs as cultural beings in their own right, a discussion about the extent to which we can simply say that cultural preferences in models mirror cultural preferences in training data, and a practical experiment with the way we make sense of model outputs as culturally aligned or misaligned at the dinner.

Kalle Kusk and Pablo Velasco ***Wasted efforts? Generatve AI in Digital Methods Education of Non-STEM students***

“I would like to scrape theguardian.com for recent mentions of the US presidential campaign. Please write the commented code in Python.” The response from ChatGPT, fully written in 25 seconds, reads “Certainly! Below is a Python code snippet that demonstrates how to scrape recent mentions…” followed by functioning code and an explanation of how it works. STS students learning digital methods have such generative AI (GenAI) tools at their fingertips, and whether they ameliorate or inhibit critical understanding rests on how they are put to use. Building on a scholarship of the political and cultural impact of new infrastructures, systems, and platforms (Marres and Gerlitz 2016, Helmond 2015, Dieter et al. 2019), this paper reflects on the emergence of GenAI tools and their seemingly inescapable presence in teaching digital methods. We argue that educators must simultaneously be sceptic of GenAI and consider how to pedagogically engage with the tools, so the teaching remains relevant to the students and inspires a critical understanding that stimulates intervention and change in technological structures (Soon and Velasco 2023). In the paper, we present in-classroom GenAI experiences with non-STEM students and reflect on the risk that the non-STEM cohort of students who are typical within STS will view acquiring a deeper understanding of digital methods as a waste of effort.

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Open Panel C

Waste as a resource? (1)
Chair: Benjamin Lipp

Staffan Appelgren ***Failed Resource Making: Circulating Waste in the Circular Economy***

This paper discusses reductive and dissociative aspects of Circular Economy initiatives using insights from ethnographic fieldwork in the field of reuse and redesign in Sweden. In recent years, the concept of Circular Economy has gained momentum as a sustainable alternative to the linear model of production and consumption, endorsed by governments, industry, and academia for its commitment to extracting valuable resources out of worthless waste. However, studies show that its implementation oLen encounters challenges rooted in corporate efficiency leading to reduction of values, dissociation of people from resources and deferrals of costs to later cycles. Taking inspiration from posthumanism and critical heritage studies, and drawing on the classic work of Michael Thompson on Rubbish Theory (1979), this paper compares material circulation among Circular Economy initiatives and redesigners and their different approaches to the multifaceted properties (f ex cultural, historic, aesthetic) of existing things and materials. According to Thompson, all transient material culture ends up as worthless rubbish/waste that only can be redeemed to the category of durables through the work of powerful “eccentric” actors, while any (re-)ascendance to the category of transient is considered impossible. The paper explores the way that both fields provide significant departures from this theory, arguing that Circular Economy projects fail in securing crucial values in circulating materials."

Taru Lehtokunnas, Alma Onali, Niina Uusitalo and Ulla-Maija Sutinen ***Translating waste flow to knowledge through multisensory ways of knowing: A composition study experience***

The Circular Economy paradigm attempts to guide waste materials back to production streams. To keep track of the types and proportions of waste produced in society, different types of waste composition studies are carried out around the world. We conducted ethnographic fieldwork in one composition study organised by a Finnish waste management company to gain hands-on experience on the ways knowledge about waste is constructed. The data includes ethnographic observations, researcher diaries, recorded discussions, photographs and video materials. We examine how knowledge about waste is produced through multisensory means within the context of a waste composition study. Drawing on the concept of translation and earlier literature of multisensory ways of

knowing, we discuss how we learned to translate our sensory and bodily knowledge of waste. We approach the composition study as a chain of translations by multisensory practices, which turns anonymous waste masses into numerical knowledge. We identify three phases in the chain, where the ontology of waste becomes transformed: 1) sample collection, 2) determining categories and 3) weighing and quantifying. We also pay attention to the dynamics of knowing / unknowing in the process of translation. Our research contributes to the understanding on the process of translation in the context of waste as we analyse how scientific knowledge about waste as an especially sensory-intensive matter is produced through multisensory practices of translation. We propose that a composition study breaks the line between private and public spheres by framing waste as a matter of concern in a specific way.

Bård Torvetjønn Haugland ***Capturing the carbon: or, solving a paradoxical state***

Carbon dioxide has become the waste par excellence: it is produced in abundant measures that threaten planetary well-being, and yet, we continue to put more CO2 into the atmosphere. One prospective way of dealing with this waste is carbon capture and storage (CCS). Since the late 1990s, national governments have supported and promoted CCS, and the concept has since become a mainstay in the IPCC's climate mitigation scenarios. Drawing upon the literature on socio-technical imaginaries, this paper digs into the surprisingly deep roots of CCS in Norway to show how the prospect of burying waste underneath the Norwegian continental shelf is both a reflection of a long-standing social order and a way of continuing that same order.
The paper draws upon a combination of policy documents, secondary literature, and qualitative interviews to explore the Norwegian State's long-standing interest in CCS. By tracing Norwegian resource exploitation from the regulation of hydropower in the early 1900s via petroleum extraction in the 1960s and to the long-standing interest in storing captured CO2 underneath the Norwegian continental shelf, the paper also develops concept of socio-technical imaginaries: this history provides insight into the interaction between different imaginaries. This furthers our understanding of how deep-seated logics impact the state's actions, including how such logic comes into play when the state navigates conflicting imaginaries. The paper identifies four kinds of inter-imaginary interaction, which all relate to the Norwegian State's attempts at navigating the paradoxical identities of petroleum producer and climate pioneer.

Panel 7

Coping with waste and excess in energy transitions – how can STS help homeless concerns and values of certain stakeholders to find a home?
Chair: Julia Kirch Kirkegaard

With this panel, we invite you to discuss and reflect on how, by whom, and with what tools and devices energy transitions are imagined, and with what consequences for the types of concerns and values that are included or excluded in the design of energy transitions. Climate change poses a grand challenge that requires multifarious solutions. Our point of departure is that energy transitions are often boiled down to a 'purified' and 'technified' matter of deploying renewable energy in the most cost-efficient way. At the same time, notions of 'efficiency', 'optimisation' and 'flexibility' in future energy systems imply very narrow imaginations of who the future 'rational' energy user should be. While this has been behind the innovations that have made wind power cost-competitive against fossil fuels, some concerns and values for 'the social', 'everyday life' or 'climate', 'environment', and 'nature' have been marginalized, made homeless, and excluded from expert tools such as digital simulation devices, procurement schemes and living lab demonstrations, causing frictions over what should come to count and matter. We believe STS can help shed light on how some values and concerns are treated as 'efficient' and others as 'excess' or 'waste' in the design of energy transitions, as is e.g. seen in the design of renewable energy infrastructures. In so doing STS could also explore how notions of 'efficiency' frame 'wastefulness' in very narrow terms. With this panel, we invite STS scholars to help us think and intervene in these valuations and 'othering' processes through conceptual and methodological innovation.

Felipe Silva Figueiredo ***When Excess Faces Its Limits: How Energy Infrastructures are Shaping Future Imaginaries in Iceland?***

Iceland's reputation as a place of untamed nature has long been related to images of excess, particularly when it comes to resources for renewable energy production. Harnessing the power of seismic activity and glacial rivers, the country has achieved a remarkable feat - powering 100% of its grid with renewable energy. This unique environment, rich in potential energy sources, has positioned Iceland as a focal point of the so-called sustainable development. However, alongside this image of excess lies a complex sociotechnical environment. Despite boasting one of the highest rates of energy consumption per capita globally, Iceland faces challenges concerning its energy supply. The discourse of plentiful and inexpensive renewable energy has shaped political and economic agendas, driving the growth of energy-intensive industries under the guise of sustainability. Yet, recent concerns have emerged regarding constraints and shortages in the country's energy supply, pushing it towards sociotechnical limits of power

generation. This presentation, drawing from Science and Technology Studies (STS) perspectives, will examine the interplay between economic growth and the sociotechnical and environmental boundaries of renewable energy infrastructures in Iceland. Through ethnographic data and interviews with key stakeholders, including political actors, representatives from the national energy company, and climate activists, the discussion will unpack the complexities of Iceland's energopolitics (Boyer 2019). This presentation will discuss how energy infrastructures mobilizes discourses of excess in Iceland and shapes future techno-economic imaginaries in the country and beyond, within the context of the ongoing climate emergency.

Sophie Nyborg, Julia Kirch Kirkegaard, Tom Cronin, Emil Nissen *Co-creating benefits and values in an energy community - potentials for reevaluating*
energy systems

STS scholars and social practice theorists have long pointed out how energy system designs and imaginations reflect system builders' and engineers' assumptions about what counts as valuable in the system design, how future societies will be configured and what roles and responsibilities e.g. users and citizens will have in their vision (e.g. Strengers, 2013; Nyborg, 2015; Nyborg & Røpke, 2013; Kirkegaard, 2023, Callon, 1986, etc.). These assumptions are built into design tools, which marginalize alternative, non-expert concerns and valuations (Kirkegaard 2023). In this paper, we explore the potential of co-creation workshops to unpack and confront taken-for-granted valuations, in order to 're-evaluate valuations' and co-create inclusive designs and visions for the future energy system. We do so by designing, facilitating and analyzing a co-creation workshop organized with board members of an emerging energy community on the Danish island of Funen, which has the aim to elicit key values of the energy community, and which benefits it can bring the future community members. Such co-created visions of benefits and values may feed into alternative design- and valuation tools in the green energy transition, but our study may also provide knowledge on how the equipment and valuation tools of the co-creative space itself shape co-creation processes and outcomes, for instance in terms of which values are discarded in the process, and which ones remain unquestioned. In doing so, we are simultaneously confronting widespread assumptions that co-creation processes are necessarily providing inclusive and democratic solutions (e.g. Elkjær et al, 2021, Nyborg et al, 2024).

Ask Greve Johansen, Julia Kirch Kirkegaard, Peter Holm Jacobsen, Jens Iuel-Stissing *Efficiency or wasteful industrial policies? The assembly of Power-to-X in the Danish bureaucracy*

Power-to-X (PtX) is an economy of wind, sun, water, pipelines, powerlines, land, and large machines - and an economy which asks for significant state intervention, support, and risk-taking. In other words, the tenets of orthodox neoclassical economics are severely challenged and what we could call industrial policy is returning to the spotlight.
In Denmark, PtX has rapidly risen to prominence in the national energy future. PtX is supporting the electrification of the energy system while substituting fossil fuels where electricity is impractical or even wasteful. Perhaps more importantly, PtX is seen across a large number of actors in the Danish industry as a promising new industrial venture, that - while helping the state in reaching its carbon reduction commitments - will generate jobs and export of products and technologies.
From the perspective of neoclassical economics, however, promoting a specific technology to achieve decarbonization goals may be construed as wasteful. In other words, the (limited) success of economic expertise in defining the playing field for energy transition hinges on its promise to prevent wasteful use of capital and material resources, instead seeking to ensure efficient allocation.

In this paper, we ask civil servants in the Danish Energy Agency working with PtX how they, as the midwifes of new policies and regulation for PtX, imagine the tension between commitments to efficient allocation (and minimal intervention) on the one hand, and then on the other, the strong political push towards promoting and paving the way for specific technologies to succeed.

Inge Røpke, Julia Kirch Kirkegaard, Tom Cronin *Power-to-X and homeless concerns for energy sufficiency*

In the face of climate change, most countries are in the midst of transitions to decarbonize their energy system. Recently, hopes for the success of green transitions have been pinned on so-called ‘power-to-x’ (P2X) technologies. The ambition is to convert water into hydrogen through use of superfluous electrons from renewable sources. P2X bodes well for sector-coupling, as the energy of overflowing electrons can be converted, stored and transported as molecules (e.g. hydrogen), and ultimately used to synthesise artificial fuels, such as ‘green e-fuels’, to be consumed outside of the electricity sector. While seemingly solving the problem of ‘wasting’ renewable energy, PtX strategies and policies still fail to provide clear guidelines, and in the development of their approaches, actors face considerable unknowns, uncertainties, and risks. Based on document analysis and interviews, we use the case of Denmark’s PtX-strategy (2021) - set in the context of EU’s green hydrogen ambitions - to explore not only

what agendas have been driving the PtX-ambitions in Danish state bureaucracy and industry, but also aiming to detect concerns regarding "degrowth" through energy sufficiency (e.g. Zell-Ziegler et al., 2021; Toulouse et al., 2019; Lindgren et al., 2023) raised by actors outside of the sector. We demonstrate how PtX and green hydrogen ambitions are – on the one hand – driven by a techno-economic desire for cost-efficient energy transitions, supporting continued socio-economic growth (see Daggett, 2019), while – on the other hand – the failure to render concerns for energy sufficiency a 'home' may have implications for the (in)ability to imagine how future energy pathways could be otherwise.

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Amanda Obitz Mogensen *Responsible investment in the Danish Pension Sector: The value of energy*

To limit the consequences of climate change, societies need to fulfil a rapid transition from existing fossil-fuel-based energy towards low-carbon energy systems. In this context, pension companies have become major targets for contributing to the energy transition. Environmental, Social, and Governance (ESG) managers are increasingly hired to implement what is in vernacular referred to as responsible investment practices. This also entails dealing with values previously excluded from finance. This paper aims to explore the problem of valuation by examining how notions of energy investments have entered a Danish pension company, AkademikerPension. The analysis mobilises the pension company's 'Responsible Investment Policy' as a device, positioning it as an analytical entry point. It furthermore builds on ethnographic research in the Danish pension sector and unfolds how knowledge regimes direct a transition narrative in finance. Based on this, I reflect upon the homeless concerns

and values that emerge from this increasing attention to finance in the energy transition.

Daniel Nordstrand Frantzen ***Struggling with participation: Hearing processes put to the test***

The importance of public participation in promoting renewable energy technologies and advancing the green transition is widely acknowledged (Wüstenhagen et al., 2007). However, studies in STS have shown that mainstream public participation frameworks can be limiting and contentious by not acknowledging emergent alternative forms of participation (Chilvers et al., 2018). In this paper, I propose that this STS agenda can be furthered by cross-fertilizing with Boltanski & Thévenot's (2006) sociology of tests, which has strong theoretical ties to ANT, though it is seldom referred to in STS (Guggenheim & Potthast, 2012; Lemieux, 2008).

Empirically, the paper explores participation in public hearing processes for wind farms that are under planning in Denmark. I conceptualize how hearings frame participation by outlining legitimate ways of 'testing' the project under hearing while excluding other types of tests.

I describe how citizens often face challenges expressing their critiques through such legitimate tests and hence resort to alternative test formats. The paper identifies two such formats: 'potentiality tests' and 'de-legitimation tests'. In the first format, locals seek to establish that other solutions are possible and more desirable than the project under hearing. In the second format, locals repurpose the legitimate test formats as instruments for canceling the project.

Both these alternative tests point to the futility – or 'wastefulness' - of participation in the current hearing format. While the ideas conveyed in potentiality tests are not taken up, successful de-legitimation tests may lead to projects being canceled. Examining these alternative tests can suggest ways to make participation more inclusive.

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Panel 6

Computational Empowerment in Primary Education: Integrating STS Methods and Concepts
Chair: Bjarke Lindsø Andersen

Researchers have already worked on how STS can contribute to the ongoing development of computational empowerment in schools (Danholt 2021; Andersen & Tafdrup 2022; Andersen, Danholt & Ratner 2023; Andersen et al. 2023). This panel seeks contributions to further the integration of Science and Technology Studies (STS) methods and concepts to enhance computational empowerment in primary education. We are particularly interested in how STS can inform and transform the teaching of technology, fostering critical thinking and ethical understanding in young learners. The focus will be on how STS approaches can guide children in assessing the consequences and uses of technology, particularly in the context of sustainability. We aim to explore ways in which STS can make complex ideas accessible to children, helping them to become informed, responsible digital citizens. Contributions may include case studies, theoretical frameworks, or practical strategies for implementing STS principles in primary education. By bringing these ideas into the classroom, we hope to equip the next generation with the tools needed for thoughtful engagement in a technologically advanced and sustainability-conscious world.

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Oliver Alexander Tafdrup ***On the connection between security politics and computational empowerment – using STS concepts to explore questions of European security***

In recent years, the field of security politics in Europe has been significantly revitalized by two pivotal events: the Russian invasion of Ukraine and the escalating great power competition between the USA and China (Ruppert 2023). These events highlight how technology and

innovation infrastructures are becoming crucial factors in the conflicts between NATO and its adversaries. The situation in Ukraine, for example, previews a future battlefield where AI, automated kill chains, and cyberwarfare are increasingly prevalent (Ford & Hoskins 2022). Concurrently, the technological race, especially in emerging AI and quantum technologies between the USA and China, underscores the potential for significant disruptions in computational capabilities, impacting all military domains, including cyberwarfare (Teglskov & Liebetrau 2023). This paper argues that the increasingly unstable security order in Europe necessitates a greater focus on computational empowerment within the education system, bringing together security and defense politics with research and education. It posits that Science and Technology Studies (STS), through concepts like Sociotechnical Imaginaries, provide essential tools for pupils and students to engage in exploring the complex interrelations among politics, technology, and security. This approach contributes to a deeper understanding of the role of computational empowerment and how technological innovation is intertwined with visions of geopolitics and the nature of future warfare. By employing the concept of sociotechnical imaginaries (Jasanoff 2015), educators can explore how collective visions of technological and political futures shape, and are shaped by, the current trajectory of European security politics. Through this lens, the paper will highlight the importance of computational empowerment in shaping the future of European security politics and warfare, underscoring the need for an integrated approach that bridges technology, policy, and education.

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Sanne Lisborg ***Tinkering, mediation, and multistability as a didactic lens in Technology Comprehension***

I unfold how the field of STS can provide a framework to develop a didactic and pedagogical approach for teaching Technology Comprehension (TC) in Primary Education. I focus on the concepts "tinkering", "mediation", and "multistability", and how they can be applied as didactic concepts teaching TC. I propose a technology analysis founded on three didactic levels: 1) The pupils "act towards knowledge" through their embodied experiences and engagement with technologies. 2) The pupils "analyze towards knowledge" by utilizing their phenomenological first-hand experiences with technology as a starting point for a mediation analysis of how technology influences their options for action, perception, and embodiment. 3) The pupils "fantasize towards knowledge" through discussions about which technological practices are (or could be) constructed in the interplay of technology, humans, and other materialities. I illustrate how these three didactic levels can be applied in teaching primary education with examples from natural science and social science.

Simy Kaur Gahoonia, Christopher Gad ***Prototypes all the way down: Prototyping in the teaching and development of technology comprehension***

The recent trial of technology comprehension tested 'a design approach' (Iversen et al., 2019; see also Gahoonia, 2024) to computational empowerment, and associated prototyping practice, grounded in the suggested educational, formative, democratic, and creative-constructive potentials of these methods in a Danish public-school context. We explore inflictions of prototyping in the trial programme. We empirically examine the performativity two prototypes (Suchman et al., 2002)—and suggest the proposed technology comprehension subject as a third. Arguing that the trial programme was 'prototypes all the way down', we discuss the

proliferation of a ‘prototyping culture’ (Corsín Jiménez, 2014) and develop prototyping as a key to 1) critically explore a design approach to computational empowerment and technology education; 2) make sense of the ‘under-realisation’ of a technology comprehension school subject in the wake of the trial through a critical perspective on experimentalism as governance (Marres & Stark, 2020).

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Laura Mørk Emtoft & Vibeke Schrøder ***Designing for participation and computational empowerment for all in an entangled technology comprehension classroom***

As a first stepping stone in a design based research project on inclusion and participation for all in the technology comprehension subject, we conduct a systematic review. Further, our context phase will consist of reframing the Universal Design for Learning (UDL), a framework developed in the 1990’s to overcome what had until then has been seen as individual learning disabilities. The three core pillars of the UDL design are 1. ‘Provide Multible Means of Engagement’. 2. ‘Provide Multiple Means of Representation’ and 3. ‘Multible Means of Action & Expression’ (Meyer et al., 2014). However, the framework overlooks the complexity and messiness of the classroom, as described through Fawns’ concept of ‘entangled pedagogy’ (Fawns, 2022). Based on the review and redesign of the UDL framework, the question this paper poses is: How might an STS perspective contribute to the design principles for an inclusive technology comprehension classroom, offering computational empowerment for all?"

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Camilla Balslev Nielsen ***Reimagining Learning Materials: A Socio-Material Approach for Computational Empowerment***

The article argues for adopting a socio-material perspective informed by Science and Technology Studies (STS) when designing and understanding technology comprehension learning materials, given the necessity of adaptation to a dynamic and uncertain world.

It begins by introducing Bruno Latour's concepts of 'hybridity' and 'translation' to grasp the complexity of relationships shaped by continuous creation and negotiation. The article unfolds the construction of the learning material "Super Animal" (translated: Superdyr) from the Danish project ultra:bit, which has seen enrollment from 9 out of 10 public schools. This process is informed by insights from Estrid Sørensen and Lucy Suchman. The creators of the learning material have solidified certain assumptions about students, teachers, learning environments, and technologies. The analysis employs Andrew Pickering's concept of the 'dance of agency' and Madeleine Akrich's notion of 'inscription' to reveal socio-material patterns, proposing adjustable principles for creating comprehension learning materials for grade school students.

In 2018, the Danish Broadcasting Corporation (DR) initiated the five-year development project ultra:bit in collaboration with The Danish Center for Teaching Resources (CFU) and The National Center for Science Education (Astra), supported by funding from the Danish Industry Foundation. Ultra:bit draws inspiration from BBC's "Make it digital" initiatives, sharing a vision of inspiring a new generation to engage creatively with coding, programming, and digital technology.

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Thea Sofie Skjødt Engstrøm ***Computational Empowerment as Ontological Choreography: Using STS Concepts to Navigate Sociotechnical Classrooms***

In this paper, I discuss the transformative impact of technology on the challenges that educators face in guiding children to become competent digital citizens within a complex digital landscape. Teachers are increasingly expected to cultivate forms of "digital literacy [teknologiforståelse] in primary schools to prepare children for a digital future, fostering an understanding and competencies in managing technology's opportunities and challenges" (Regeringen et al., own translation). It, therefore, seems crucial to understand how educators simultaneously deploy notions of digital literacy in the creation of learning environments while ensuring that the various elements of computational empowerment, such as emerging technologies, digital confidence, and the confrontation of cultural stereotypes associated with digital technology, is achievable for all students. Here, I suggest that the concept of ontological choreography (Thompson, 2005; Cussins, 1996) is particularly useful for unpacking and inhabiting the intricacies of digital technologies as sociotechnical systems (Seaver, 2017), thereby deepening educators' understanding of their central role in nurturing computational empowerment. The goal is to help teachers more easily navigate the sociotechnical complexities that influence students' digital engagement. I argue that by adopting Thompson's framework, educators can dissect and address the choreographic coordination of multiple and ontologically distinct elements that constitute classroom dynamics, fostering an environment where computational empowerment extends beyond digital literacy, integrating into students' identities and enabling them to tackle technological challenges effectively.

Panel 9

Waste Legalities of Affluence and Effluence (workshop)
Chair: Michael Hennessy Picard
Co-conveners: Leonora Kleppa Stærfeldt, Garance Thomas, Mikołaj Szafrański, Michael Picard

Inspired by Sheila Jasanoff's concepts of "sociotechnical imaginary" and "co-production", the panel will review current Waste Regulation & Policy through the lenses of STS. The "sociotechnical imaginary" of waste governance is defined as the imagined forms of social order centered on the scientific and technological management of the waste problem. The sociotechnical imaginary of waste management is the belief in the capacity of circular reuse, recycling and refurbishment to achieve sustainability, without fundamentally engaging with the toxic legacies of the global waste economy. Such a sociotechnical imaginary suggests that new technologies of recycling and repair will redesign key aspects of the global economy and optimize waste management infrastructure, while customs monitoring through the systematic use of image recognition, robotics, remote sensing and scanning will improve illegal waste shipment detection.

But what if the new clothes of this recycling empire were hiding the larger effluence generated by the global economy? If electronic circuits are now "salvaged" by the global multi-billion-dollar electronic waste-recovery networks, their sociotechnical imaginary also co-produce covert categories of mining waste, as well as negative fractions which in many countries remain "outside" of the circular design of the Green New Deal.[1] In the waste management economy, invisibility, opacity, illegalities and uncertainty persist despite legal developments of transparency, monitoring and extended producer responsibility. As the positive pole of formal and licit waste management expands, so does, it seems, the negative pole of informal and illicit trade.

If visibility of waste in the circular economy co-produces invisibility of waste –extractive, human, etc. –, how does legal scholarship account for this sociotechnical imaginary? What tools can legal scholarship mobilize to engage with the co-production of affluence (value) and effluence (leakage)? What kinds of legal techniques facilitate the techno-solutionism of the circular economy? How does law make it possible to classify and establish value, reifying certain charismatic waste objects and making others invisible? What dreamscape of legal jurisprudence can we propose to engage in an 'ecological' relationship with objects, thus preventing their production? To discuss these themes, our workshop will offer a structured conversation between four legal scholars: Leonora Kleppa Stærfeldt (University of Copenhagen Faculty of Law), Garance Thomas (Sciences Po

Law School), Mikołaj Szafrański (LSE Law School) & Michael Picard (University of Edinburgh Law School).

Panel 8

Data (after)lives in the digitalized welfare state: Exploring Gaps, Frictions and Surplus
Chair: Helene Ratner

The digitalization of the Danish welfare state, transforming from human interactions and paper-based archives to computer-readable formats, has reshaped the relationship between the state and its citizens. In the realm of digital data, citizen information is not merely stored; it is produced, aggregated, circulated, combined, and repurposed in novel ways.

The state increasingly views this wealth of data as a strategic resource. Central in digitalization narratives is the projection a future where Denmark leads Europe in utilizing data and artificial intelligence. As a surplus value, data are anticipated to mitigate public resource consumption, alleviate workforce shortages in welfare sectors, and support the green transition. However, as numerous Science and Technology Studies (STS) have revealed, the path to generating and leveraging data is fraught with frictions, uncertainty, gaps, and unintended consequences (Edwards et al., 2011; Gitelman and Jackson, 2013; Jaton, 2017).

This is evident, for instance, in the production of data. Today, citizens navigate digital platforms for tasks ranging from passport applications to social benefit requests. Herein lies a new form of datafication, where citizens are cast as data providers, responsible for furnishing online systems with accurate and requisite information. However, data gaps and friction emerge when citizens, lacking digital access, cannot take on the role as data providers.

Data surplus, conversely, is produced as the state starts to perceive vast digital archives as reservoirs of 'surplus value,' envisioning their utility in developing AI technologies. This repurposing includes leveraging citizen data for machine learning processes, used to develop new profiling and predictive tools for future welfare provision (Ratner and Elmholdt, 2023). In this context, citizen data, once collected for making decisions about the citizens in question, acquire new data ‘afterlives’ in their transformation to algorithmic rules about how to view other citizens (Ebeling, 2022).

In this panel, we are interested in examining the frictions and gaps becoming manifest in the digitalized welfare state’s multifaceted landscape of data lives and afterlives: From the production of digital data over its processing to its reuse in new contexts.

Brit Ross Winthereik ***Life as surplus in digital welfare service systems***

The paper explores the evolving role of citizens in digitalized welfare services and highlights the limitations of their agency in addressing unexpected events within these systems. While citizens contribute data and manage their affairs online, they often rely on informal helpers to address

problems or escalate issues to authorities. The narrative emphasizes the need for joint collaboration between citizens and frontline personnel to identify flaws and improve processes in digital service systems.
Critical perspectives on 'the digital' reveal concerns about the reliance on private actors, such as banks, in providing access to welfare services. While private companies play a significant role in digitalization, they may not be held accountable for access issues in the same way as public authorities. This raises questions about ensuring inclusivity and avoiding exclusion in digital ecologies with private street-level bureaucrats. The paper advocates for a shift towards a perspective of repair and learning within digital systems, emphasizing the importance of institutional preparedness and responsiveness to citizen experiences. It suggests fostering empathy and attunement during encounters with citizens, as well as monitoring their experiences of exclusion to inform systemic improvements.
Overall, the paper calls for a reevaluation of digitalization efforts to prioritize inclusivity and address the complex challenges of welfare service delivery in digitalized societies.

Frederik Schade ***State Digitalization and the Changing Praxis of Data Destruction in Public Bureaucracies***

If the production of knowledge constitutes a key characteristic of institutions, so too does its destruction. In the context of democratic states, this dynamic is exemplified by regulatory norms concerning the (selective) preservation and potential destruction of public records for reasons of both government accountability and cultural memory. Rather than giving rise to public controversy, the continuous destruction(s) of information guided by such simultaneously legal, bureaucratic, and archival regimes constitute a deeply institutionalized element of bureaucratic praxis. However, as both the practices of government and the communication of public officials increasingly rely on digital technologies and platforms, mundane forms and procedures of data destruction in the public sector are being simultaneously challenged, transformed, and re-evaluated. Digitally enabled correspondence (e.g., emailing), ephemeral messaging facilitated by private platforms (e.g., the case of “government by Whatsapp”), and emergent concerns around data privacy thus all – in different ways – unsettle pre-existing norms for administering public records and other types of government data. Exploring these normative and practical disruptions, this presentation lays out a framework for the ethnographic study of mundane practices of data destruction in public bureaucracies. In particular, it centers on transformations in how data destruction is

practiced and valued within increasingly digitalized democracies and the role of technological affordances in shaping these transformations. From these developments, questions emerge pertaining to both cultural memory and political accountability in the wake of digitalization.

Torben Elgaard Jensen, Alfred Lund Felumb ***Covid Management and Its Consequences: Preliminary analyses from a field study at the Danish Patient Safety Authority***

Organizations do not digitize once and for all, but rather through an incessant stream of data projects motivated by a variety of agendas and events. The aim of this paper is to describe and discuss the longer trends in how a particular Danish authority, Styrelsen for Patient Sikkerhed (STPS, the Danish Patient Safety Authority), has been digitizing its operations over the past five years.

The STPS is a particularly interesting case of digitization because the authority has been subject to a variety of political agendas, ambitions, and events that might potentially shape its digitization. Several potential influences could be listed: (a) The long-term pressures on the public sector to use digitization as a tool for modernizing/effectivizing (Schou & Hjelholt 2019), (b) The hopes associated with 'big data' and the associated ambitions of gaining new kinds of value by harvesting and analyzing internal and external data in novel ways (Mayer-Schönberger & Cukier 2013) (c) The idea, sometimes referred to as 'the digital shock doctrine' (Klein 2020) that historical crisis events such as STPS's key role in the COVID management will pave the way for radically new and perhaps less precautionary ways of using data.

The presentation analyzes the trends in the digitization of the STPS in light of the potential influences listed above. The analysis is based on a two-month ethnographic field study of the STPS, including a comprehensive analysis of documents describing STPS's internal data projects.

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Ida Schrøder, Jacob Laage-Thomsen, Helena Ratner *An emerging welfare data economy?*

> *"*This is the cool thing about this project. We can examine what is possible and where the limits are. There is the business value – and then there's the technical aspect, that we become confident with (…) [machine learning] more generally. And then we can implement this in other places*"* (Anne, Project Manager in a Danish municipality).

In this paper we explore how business ideas of data as assets (Birch et al., 2021) are entwined and transformed in what we term an emerging 'welfare data economy' of the Danish Welfare state. By naming it the 'welfare data economy' we aim to take seriously the assumption that data can become assets for the welfare state. However, as Anne's quote reminds us, this is an emerging and unsettled economy, testing and experimenting with the possibility that citizens' data might be exchangeable for non-economic forms of value (Asdal et al., 2021). Tracing how the Danish authorities' changing procedures for demarcating and ordering the values of attempts at repurposing statutory citizen data, we suggest that a welfare data economy is growing from business ideas of investing for future returns. More specifically, we probe into how the demarcation and ordering of possible values unfold in the material manifestations of such a welfare data economy.

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Tobias Pedersen *Denmark's role in transnational digitalization: Expertise, knowledge travel and friction across borders*

Denmark has long been recognized as a frontrunner in digitalization. However, researchers have argued that since the turn of the 2000's, Danish public digitalization policies have gradually moved away from focusing on existing welfare logics and values, towards policies relying on international competition, flexibility and optimization (Schou and Hjelholt 2018).

Concepts such as safeguarding the weakest citizens and participatory democracy have gradually been replaced, where citizens are increasingly framed as 'self-responsibilized' and digital technologies reframed as means of cutting expenditure and securing optimization of governmental labor processes (Schou and Hjelholt 2018). These dynamics relating to the 'competition state' are highly characterized by transnational collaborations (Schou and Hjelholt 2018). Denmark is increasingly taking a leading role in these collaborations, framing itself as a model for inspiration and 'exporting' best practices of public digitalization through international forums such as Digital Nations and Digital Hub Denmark. In this regard, it becomes highly relevant to investigate how public digitalization expertise is ascribed value in the everyday practice of digitalization professionals when expertise and knowledge travel across borders. Through ethnographic fieldwork situated at Digital Hub, and a theoretic framework of Science and Technology Studies, this article investigates moments of friction (Tsing 2005) and infrastructural complexities (Bowker and Star 2000; Winthereik and Wahlberg 2022) between international communities of practice in the translation and standardization (Star 1990) of best practices through transnational digitalization.

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Open Panel E

The design of everyday life, systems, practices
Chair: Sophie Nyborg

Emma Atkins ***Did food waste concerns affect the evolution of household fridge design in the mid-20th century?***

Food waste is an enormous challenge, and it is becoming ever-more apparent that the refrigerator plays a key role in household food waste. Food is often hidden and forgotten inside, and as a significant object in daily food consumption which constrains some practices and allows others, the design of the fridge warrants closer attention. The fridge as we know it was invented in the USA in 1927, and while advances were made in energy efficiency, insulation and choice of refrigerants, for the next almost 100 years, designers have been reluctant to shift away from the familiar white box that lives in our kitchens. Although fridges were initially designed to prevent food spoilage, visibility issues and increased storage space arguably created food waste too. By analysing American fridge adverts from the 1920s to the 1960s, this research explores how 'food waste' was conceptualised in relation to the fridge during its evolution, what problems fridge companies claimed their designs would solve, and how fridges related to the wider consumption of food. Much of STS scholarship is concerned with paths not taken, and this presentation also uses archival material to investigate a case study of a 1954 General Electric model that professed to solve issues of food waste within the fridge itself – but did not 'stabilize' and shortly ceased production. Understanding the evolution of this technology, why some versions fell by the wayside and how its materiality affects waste practices today, will help design a fridge suitable for the challenges we face ahead.

Vladimir Alejandro Ariza-Montañez ***Navigating time and Nutrition: Insights from Supplementary and High-Nutrient Foods in Colombia's 1000 Days Program.***

In Colombia, different local studies and national surveys has identified the increasing prevalence of conditions related to low birth weight and acute malnutrition among children, particularly in rural and remote areas. These challenges disproportionately affected ethnic communities, exacerbating existing disparities in health outcomes. In response to this pressing issue, the Colombian government has implemented various care strategies, focusing on early identification and outpatient management of malnutrition cases, promotion of healthy lifestyles, and the distribution of supplementary food. Of note is the government's emphasis on the design and distribution of modified and fortified foods containing essential

minerals and nutrients, aligning with a trend identified by Alice Street as "nutriceuticals" (Street, 2015) within humanitarian nutrition. As part of a nutritional recovery program titled 1.000 Dias Para Cambiar el Mundo (One Thousand Days), two novel Supplementary Foods have been introduced: "Vitalac" and "Bienestarina Mama". This analysis delves into the narratives and the "humanitarian imagination" (Redfield, 2012) surrounding these foods, which cater to two timeframes: firstly, the immediate need to respond to emergencies and ensure survival; and secondly, the crucial task of supporting early childhood development, especially during the formative "window of opportunity" in the first two years of life.

Laetitia Forst, Nettie Abbot, Sarah Key, Alberto Iranzo, Sarah Gray, Kate Goldsworthy, Tim Cross, Jenny Valarino, Padouk Fielding
Measuring the potential environmental benefits of design for recycling in fashion: from design research to carbon calculations

To address the issue of waste in fashion and contribute to resolving the industry's impact on the environment, the relationship between the design stage of textiles and their treatment at end of life has become increasingly important. As global policy and recycling technology progress, the suitability of end-of-life garments as feedstock for textile-to-textile recycling becomes critical. The way garments are designed today must change so that they can be efficiently recycled tomorrow. To support the take up of design for recycling across fashion brands and retailers, measurements of potential environmental benefits can be a key leverage point to convince decision makers. This paper studies how the environmental benefits of design for recycling can be calculated and explores the potential impact and benefits of a system where as many textiles as possible are designed in this way.

Working collaboratively with the Circular Textiles Foundation, the University of the Arts London and WRAP have developed a simplified method connecting data available to brands with carbon calculation approaches derived from Life Cycle Assessment. The methodology classifies products against their fibre type and level of disruptors to recycling, providing an overview of which garments can be recycled and how. This estimate can make the case for investment in circular design.

The paper will lay out the approach used to connect design research and environmental science, layering the methods specific to each discipline, and investigating the field of commercial fashion design practice. The

resulting calculations will be presented in the context of this interdisciplinary collaboration.

Kiersten Thamm and Georgina McDowall ***Bewildering Things and Material Legacies: Recontextualizing Quotidien Objects in Stories from a Likely Future***

Bewildering Things and Material Legacies, our current project, is a climate-fiction multimodal book about the impact of everyday design on people and places. It begins with a novella that follows the book's protagonist, a researcher named Kit, as she discovers a cache of objects in 2485 sitting in the hull of the wrecked HMS Prowess. Through this story, readers learn about her environment and society, which we developed as science-informed approximations of the reader's future. A series of Kit's research entries follow the novella to bring overlooked aspects of normal objects to the fore. We imagine her researching — without the guidance of documentation or archives — by using, misusing, imagining, and experimenting with the objects themselves to understand how the ancestor-reader used and valued them. Her confusion and questions highlight how these seemingly innocuous designs not only impact the environment but intertwine with social themes including global connections, gender and sexuality, disability and access, colonialism and racism, safety and health, labour conditions, and materials extraction. Ultimately, we hope readers develop a new understanding of the quotidian objects she discovers and how their production, use, and obsolescence impact communities and ecologies.

Panel 10

Long-term waste, Slow disasters
Chair: Nina Toudal Jessen and Sebastian Lundsteen

Waste is considered ubiquitous and an inevitable outcome of living in "Late Industrialism." (Fortun 2014). The cumulative outcomes of societies informed by ideas of growth have manifested in multiple ways thus not always readily visible to the eye. Waste depends on systems and infrastructures that remove unwanted or discarded materials in order to make problems of waste invisible and the system(s) seem coherent (Liboiron & Lepawsky 2022). While waste may seem inevitable in modern life, managing and administrating waste is both politically and socially contested and opens up questions of responsibility and expertise. The seemingly innocent packaging is connected to larger systems where economic ideas or technological advantages do not always provide adequate answers (or solutions) to a booming waste economy. Moreover, as historians have shown, global waste trade is deeply implicated in colonial and capitalist structures of exploitation (Müller etc). "The future of plastic is in the garbage can" the quote that STS Metis scholar Max Liboiron uses to remind us that disposability is not only produced and intimately connected to increasing consumption, and also suggests that waste has a history (Liboiron 2021). Therefore, this panel addresses the long histories of waste and excess through the 20th century. Inspired by Joseph Masco who argues that the crisis in crisis (also waste crises) stems from the failure to examine the historical conditions and thus the roots of the crises, we invite proposals that analyze waste as temporal phenomena with a past, a present, and a future.

Nina Toudal Jessen ***Waste sites as landscape elements.***

In cultural geography and planning, a landscape analysis contains a list of landscape elements. These shape and characterize a place, thereby making it distinct. It may be specific trees, open fields, or certain views. Technical sites, such as power stations or landfills are not included as landscape elements – a category reserved for "naturally" occurring objects. However, in post- WWII Denmark, landfills were indeed a landscape element. Scattered throughout Denmark, approximately 1200 local landfills dotted the landscape, often hidden in former gravel pits or bogs. While this practice declined after the invigoration of environmental laws in the 1970s, the sites themselves still exist as the materials dumped there still deteriorate, and polluted run-off now reach the ground water basins. The contamination has reopened a discussion on whether and how to act against the chemical pollutants hidden in bogs and gravel pits. However, public calls for more control and knowledge about former waste sites echo past calls. In this presentation, I will shed light on the knowledge processes surrounding waste site management in the past, and ask how waste sites seemingly disappear from public eye until new types of measurements, new legislation or new thresholds reinstate them in the public debate in a seemingly cyclical movement.

Louise Skyggebjerg ***From ban on cans to undisputed packaging – the story of the 'Can War' and the Danish recycling system for beverage containers***

This paper adds a new materiality perspective to the literature on environmental politics focusing on the role of business, thus synthesizing two scholarly conversations. It presents a historical case study of the struggle over a Danish ban on metal cans for beverages discussed since 1970. The focal point is 'The Can War', a conflict between Denmark and the EEC/EU from 1980 to 2002 over free trade contra environmental protection. It involved among others Danish brewers with different and changing interests, foreign brewers claiming the ban to be an illegal trade barrier, the packaging industry, and the retailers managing container returns. By its dual examination of business motivations and materiality, the paper traces a complex set of business responses to environmentalism and show the importance of materials and infrastructures. Thus, the case demonstrates the complexities of environmental politics and the many actors – humans and non-humans – that influence packaging and wasting practices.

Rosa Nan von Leunbach ***Long lived waste, short lived solutions: The past, present and future of radioactive waste management in Denmark***

The historical narrative of radioactive waste management reveals critical issues of infrastructure, visibility and the intricate interplay between human, political and material temporalities. This discrepancy, the fluctuating cycles of political agendas, and the enduring hazard of radioactive materials, sets the stage for a complex discussion on temporal dynamics and waste.
Building on prior research (Leunbach & Nielsen 2020), this paper explores the past, present and envisioned future of the Danish radioactive waste. In the techno- and growth-optimistic atomic age post World War II, waste challenges were somewhat overshadowed by the promises of a new form of energy, when the Danish nuclear research facility, Risø, was established. However, as political and public sentiment shifted away from nuclear power and nuclear energy was excluded from the national energy mix, the waste was tucked away in Risø's temporary facilities until 2003. The decision to decommission and find a permanent solution brought the issue back into the public eye.
The subsequent 15-year socio-technical controversy, characterized by dynamic interactions in a "hybrid forum" (Callon et al., 2009), showcased the interplay between municipalities, protest groups and public experts.

The forum approach redirected decision-making from a final repository to temporary storage at Risø, postponing a permanent solution. This extended process illustrates how public deliberation in hybrid forums reshapes long-term environmental challenges.

In my paper, I will delve into the evolving waste infrastructures during the history of Risø. Insights into political processes and the public debate of radioactive waste management in Denmark will highlight the ongoing challenges posed by the intersection of temporalities in environmental stewardship and public policy.

Arts-based explorative panel

Arts-based explorative panel: Never ending endings – embracing decay as radical sustainability
Chair: Martin Grünfeld
Co-conveners: Amalie Schjøtt-Wieth, Eduardo Abrantes, Martin Grünfeld, Tim Flohr Sørensen, Louise Whiteley

Endings are unsettling. They bring to mind uncertainty, loss, and death. But what if we think about endings as never ending, and embrace decomposition as a chance to learn, create, and wonder, rather than mourn? We live in a time haunted by the remnants of overproduction and our massive consumption of things. Do we need to rethink our relationship with decay? In this alternative format, we will mix an interdisciplinary academic panel discussion with the screening of a short film and the exhibition of an installation made of waste, wax worms and other organisms. The alternative panel is based on a transdisciplinary experimental project that took place at Medical Museion between 2020 and 2023 – *The Living Room* where discarded museum objects were encouraged to decay. Museums are usually meant to preserve objects and knowledge, pristine and unchanged by time. But all objects decay, and museums cannot preserve everything forever. *The Living Room* used artistic interventions to explore what happens when endings evolve. One was "Worm Dome" – a sealed glass container where wax worms digested the plastic of medical laboratory waste, which then became a decaying ecosystem full of new creatures. In 2023 the dome was smashed open and excavated – as shown in the film "Weird Excavation". In this alternative format panel, we will screen the film "Weird Excavation" and exhibit the remains of the installation "Worm dome". The film and the installation will act as material and artistic dialogue partners for the project's makers (an artist, a conservator, a curator, a philosopher and an archeologist) to develop their thoughts about remains and remnants – what to do with excess and the left overs? How can we responsibly care for endings when decay has so many environmental impacts? And can finding ways of transforming old medical waste through art open up new conversations about sustainability, contamination, and our interdependence with the world?

Panel 8 session 2

Data (after)lives in the digitalized welfare state: Exploring Gaps, Frictions and Surplus
Chair: Irina Papazu

The digitalization of the Danish welfare state, transforming from human interactions and paper-based archives to computer-readable formats, has reshaped the relationship between the state and its citizens. In the realm of digital data, citizen information is not merely stored; it is produced, aggregated, circulated, combined, and repurposed in novel ways.

The state increasingly views this wealth of data as a strategic resource. Central in digitalization narratives is the projection a future where Denmark leads Europe in utilizing data and artificial intelligence. As a surplus value, data are anticipated to mitigate public resource consumption, alleviate workforce shortages in welfare sectors, and support the green transition. However, as numerous Science and Technology Studies (STS) have revealed, the path to generating and leveraging data is fraught with frictions, uncertainty, gaps, and unintended consequences (Edwards et al., 2011; Gitelman and Jackson, 2013; Jaton, 2017).

This is evident, for instance, in the production of data. Today, citizens navigate digital platforms for tasks ranging from passport applications to social benefit requests. Herein lies a new form of datafication, where citizens are cast as data providers, responsible for furnishing online systems with accurate and requisite information. However, data gaps and friction emerge when citizens, lacking digital access, cannot take on the role as data providers.

Data surplus, conversely, is produced as the state starts to perceive vast digital archives as reservoirs of 'surplus value,' envisioning their utility in developing AI technologies. This repurposing includes leveraging citizen data for machine learning processes, used to develop new profiling and predictive tools for future welfare provision (Ratner and Elmholdt, 2023). In this context, citizen data, once collected for making decisions about the citizens in question, acquire new data ‘afterlives’ in their transformation to algorithmic rules about how to view other citizens (Ebeling, 2022).

In this panel, we are interested in examining the frictions and gaps becoming manifest in the digitalized welfare state’s multifaceted landscape of data lives and afterlives: From the production of digital data over its processing to its reuse in new contexts.

Helene Ratner ***Failed resuscitation of welfare data: Gaps between AI promises and municipal data lifecycles***

Promises about how AI can advance Danish welfare abound, outlining potential breakthroughs in personalized service provision and a more efficient resource allocation. In realizing these promises, municipal

administrative and case work data are increasingly thematized as ‘assets’ . Using machine learning to learn and generalize from patterns found in welfare data, data bearing the traces of past State-citizen interactions are to gain new ‘afterlives’ as predictive algorithms that can sort, categorize, and target the population in novel ways (Ebeling, 2022). This paper examines one municipality’s practical efforts in implementing an algorithm for anticipating citizens’ profit from an online rehabilitation training program. To do so, the municipality needed to re-train the algorithm on its own municipal data as well as ensure its legality. The paper examines how the project members navigate various the frictions and tensions that resulted from this ambition, ultimately abandoning the project. Concretely, the paper analyzes frictions encountered with the municipality’s IT system, the need to anonymize data, and legislative uncertainties. Each forming as a distinct strategy for breathing new life into welfare data, the paper illuminates the gaps and interstices between AI promises and municipal data lifecycles.

Patrick Heiberg Kapsch & Peter Danholt ***Data Moments: A Study of the Sociotechnical Practices in Datafication of Placed Vulnerable Children***

This presentation focuses on the digitalisation and datafication occurring at a residential care centre in Aarhus, Denmark. The study explores how the daily lives of vulnerable children are intertwined with datafication. Based on ethnographic fieldwork, we propose that the datafication of vulnerable children can be studied and conceptualised as 'data moments'. These moments are mundane socio-technical practices incorporated into the daily rhythms of specialised care work that turn specific aspects of social interventions – including the child's behaviour, well-being and progress – into data. Conversely, they describe how digital systems and documentation render actions and practices into data-capturable formats. The presentation thus aims to contribute to research on datafication by introducing a concept that sensitises us to the socio-technical practices of datafication.

The analysis unpacks the data moments identified in the fieldwork, showing how the datafication of placed vulnerable children is shaped by different pedagogical tools and caregiving approaches, as well as various IT systems, analogue tools, law, and institutional frameworks that mandate frequent evaluation and quantification of the child's progress based on action plans and formal objectives. We identify the circular dynamics of datafication at the care centre. Lastly, we discuss how datafication supports care work, e.g., how the datafication of vulnerable children emerges as a productive lever for the caregivers' daily work and how new quantification practices generate specific data imaginaries.

Christoffer Koch Andersen ***Data Deathworlds and Digital Trans Liveability: The Data Haunting and Data (Non)Lives of Trans Subjects in the Digitalised Danish Welfare State***

The digitalisation of the Danish welfare state has promised to make our lives and access to state services 'efficient' and 'effortless' – from governing taxes and salary payments, bank account logins, healthcare appointments and passport applications to MitID-verification that construct our datafied citizenship. What occurs when this promise of 'optimised' digital services and compulsory datafied citizenship are only reserved for cisgender citizens? What happens when instead of creating thriving data lives, the digitalisation excludes non-cisgendered subjects from participation and creates data 'deathworlds' (Mbembe, 2003), where trans citizens find themselves either *haunted by their data* (Mackenzie, 2017) or *trapped in a state of data non-living* (cárdenas, 2017); unable to be read by the digital state (Figure 1.1) and 'perform' their data citizenship or access vital functions of the welfare state?



Figure 1.1. Network of Digitalised Danish State Services and Trans Incomprehensibility. 2

This paper investigates the necropolitical frictions within Danish data lives and the precarity of trans citizens in the digitalised welfare state by

examining the relationship between trans citizenship and their data (non)lives under the binary gaze of the digital Danish state. Firstly, I look at 'Data Haunting', where past data created about trans people do not disappear upon legal gender change; rather, it intimately haunts them and their data in the present. Secondly, I attend to 'Data (Non)Lives' to show how binary state logics of gendered CPR-numbers and binary algorithms underpinning the digitalisation create a toxic non-existence for trans citizens, where they become trapped in a state of 'non-living' as they are excluded from digital welfare services.

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Louis Ravn, Nanna Bonde Thylstrup *The legal, infrastructural, and political frictions of digital data afterlives in the Danish National Archives*

As part of the digitalization of Denmark and its collective memory, the Danish National Archives (Rigsarkivet) is responsible for the archiving of an increasing amount of digital data. Among other things, this entails the storage of various types of data which the National Archives receive from Danish municipalities and regions. This paper approaches the processes by which the Danish National Archives store these digital data not as linearly accumulative, but instead as fraught with legal, infrastructural, and political frictions that condition which data are preserved, and which are disposed of. Drawing on policy documents, institutional strategies, and interviews, this paper analyzes three frictions inherent to the preservation and destruction of digital data in the Danish National Archives: first, negotiations around which data are designated as "worthy of preservation" and which ones as disposable; second, tensions that emanate from relying on corporate data storage infrastructures; and third, the political dimensions of data transfer across different levels of bureaucracy. In so doing, the paper sheds light on how the afterlives – and deaths – of digital data in the Danish National Archives are shaped by complex entwinements of legal frameworks, infrastructural conditions, and political negotiations. The lifecycles of the digital data stored in the Danish National Archives, the paper concludes, are

contingent upon the idiosyncractic interplay of legal, infrastructural, and political conditions.

Barbara Nino Carreras *On Being Stopped by Digital Welfare: Institutional Mechanics of Hostility and Inefficiency in Denmark*

Government authorities across the world are implementing digital technologies to govern populations by promising security, efficiency, and convenience to its citizens. Human rights organizations and other civil society groups, however, warn that digital technologies delegate increased responsibility and the labor of access to citizens and informal networks of care and mutual aid. Furthermore, as technologies are designed to mediate the interaction between citizens and the state, institutional forms of ignorance and injustice become tangible when efficient processes and the benefits of digital technologies are only experienced by those already in positions of privilege and power. Drawing on an ethnographic study in Denmark in which I collaborated with with low income, racialized, and disabled citizens, I use graphic stories and image descriptions reflecting on the affective (Ahmed 2004) , temporal (Kupper 2014), and political implications of *the institutional mechanics of being stopped (or not) by digital welfare provision*. Drawing on Sarah Ahmed's work (2012; 2021), these multimodal ethnographic stories critically examine how digital welfare infrastructures, policies and practices, like complaint processes, can lead to strategic inefficiency, and hostility (Narita 2023). Strategic inefficiency and hostility materialize as vulnerable populations experience that accessing their own money or entitled benefits is more difficult in a digital society. Or when individuals and collectives are othered by digital infrastructures and inclusion initiatives as problematic subjects, whilst their access needs and knowledges are ignored or downplayed. Through multimodal ethnographic stories, I contribute to this panel by exploring the affective, temporal, and political dimensions of digital-by-default welfare provision in Denmark beyond questions of data extraction and use. I center the importance of scholarly and policy discussions that reckon with the unequal distribution of power and privilege in processes of digital transformation.

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